

Shunting Inhibition and Dendritic Branching Shape Local Credit Assignment

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Abstract

Biological neurons assign credit through structures that standard neural networks usually ignore: synapses are distributed across branching dendrites, excitatory and inhibitory inputs evoke synaptic conductances, and inhibition can shunt local voltage by changing total input resistance. We study conductance-based dendritic networks with E/I synapse banks, shunting inhibition, and tree-structured branch-to-soma coupling, and ask when restricted somatic feedback can approximate compartment-specific backpropagated errors. Exact gradients for these dendritic trees have the form $\text{gradient} = \text{local eligibility} \times \text{compartment error}$: the eligibility term uses presynaptic activity, driving force, and input resistance, while the sole fast non-local term is a path-specific compartment error. Under identity upward transfer, that error is the soma-specific error transported through conductance-stage path gains. This factorization turns local learning into a credit-signal compression problem. Shunting is useful when it reshapes the path-specific error field so that the available feedback object—global scalar, neuron-wise/per-soma, low-rank, or path-structured—is a better approximation, not because it makes LocalCA a close copy of backpropagation. We test this mechanism across exact-gradient reconstruction, oracle transport, inhibition intervention, competence, morphology, and harder-data diagnostics. Shunting reduces gradient-scale mismatch, lowers intrinsic exact-error rank in the regimes studied here, and improves direct broadcast-fidelity diagnostics relative to matched additive controls. Post-training inhibitory-conductance interventions show that learned inhibition is sample-dependent and functionally important to the trained forward computation, while transported-error oracle experiments show that restoring the exact effective path-gain error substantially reduces the local-learning gap. Under nonnegative conductances and neuron-wise/per-soma 5F feedback, shunting LocalCA remains 5–6 percentage points below matched backpropagation on MNIST, Fashion-MNIST, and figure-ground MNIST. These results suggest that E/I conductance, shunting inhibition, and dendritic branching can reshape credit-signal geometry and make restricted local learning more faithful in the regimes studied.

1 Introduction

Unlike point units in standard neural networks, biological neurons assign credit across branching dendritic trees. A synapse does not only see presynaptic activity and a scalar postsynaptic activation; it also has access to local membrane voltage, synaptic reversal potential, total conductance, and input resistance. Inhibitory synapses can therefore do more than subtract activity: by increasing total local conductance, inhibition can shunt voltage and change the gain of every dendritic path passing through that compartment.

We examine whether these biophysical ingredients can facilitate local credit assignment (LocalCA). Backpropagation [15] would assign a distinct error to every dendritic compartment, whereas a biologically plausible supervisory signal is more likely to arrive as a low-bandwidth somatic or modulatory broadcast [20]. The central question is therefore not whether such a broadcast can reproduce unconstrained backpropagation in every setting, but when the exact dendritic error field is simple enough for it to approximate. Our contribution is not simply to add dendritic structure to a neural network, but to show how conductance-based synapses, shunting inhibition, and dendritic topology change the geometry of the credit signal itself.

Starting from conductance-based dendritic voltage equations [1], we derive exact gradients for dendritic trees (Theorem 1). Each synaptic gradient factorizes into a synapse-local eligibility term and a single non-local compartment

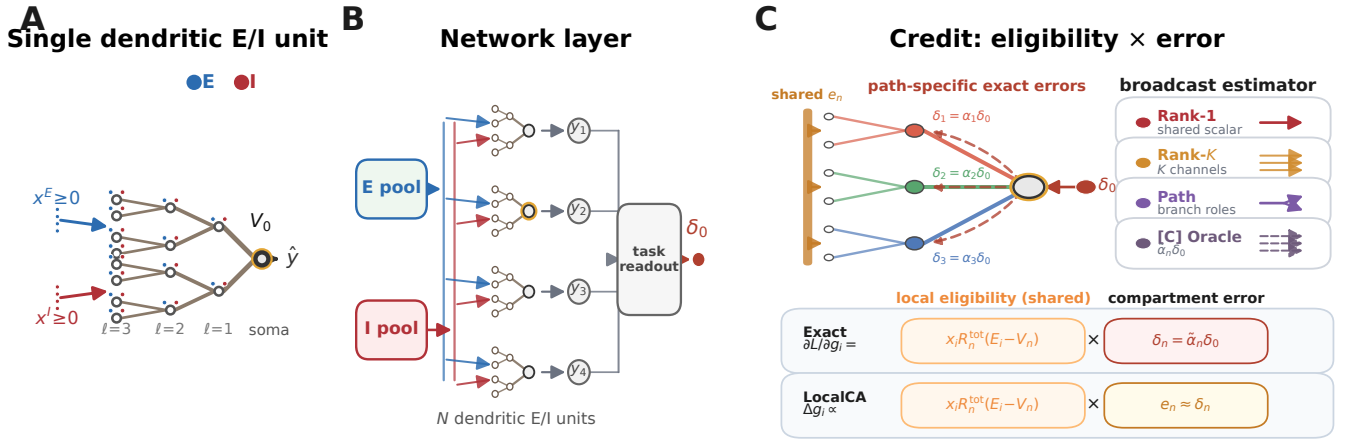


Figure 1: Model and credit assignment. (A) Single dendritic E/I unit with nonnegative excitatory and inhibitory input streams, branch synapse banks, and a gold-ringed soma matching the highlighted unit in B. (B) Layer of dendritic E/I units projecting to a task readout; $\delta_0 = \partial L / \partial V_0$ is the layer-wide soma/core error from which feedback is constructed. (C) Credit factorization: exact updates use the same local eligibility multiplied by path-specific compartment errors $\delta_{u,n}$, whereas LocalCA keeps the eligibility and replaces $\delta_{u,n}$ with a broadcast estimator $e_{u,n}$; global scalar, per-soma, Rank-K, Path, and Oracle broadcasts trade bandwidth for fidelity.

error:

$$\text{gradient} = \underbrace{\text{local eligibility}}_{\text{presynaptic drive, } E-V, R^{\text{tot}}} \times \underbrace{\text{compartment error}}_{\text{path-specific non-local term}}.$$

The compartment error is path-specific: under identity upward transfer it is the corresponding soma error multiplied by a conductance-stage path gain through that dendritic tree. LocalCA preserves the exact local eligibility and replaces this path-specific error with a broadcast field.

What is local in LocalCA? The synapse-local factor is the eligibility term: presynaptic drive, local voltage, driving force, input resistance, and optional reliability gates such as $r_n^{4F} \phi_n$. The fast task-dependent non-local quantity is the compartment error, approximated by a broadcast field e_n . Thus LocalCA variants differ by the broadcast estimator and optional reliability gates, while preserving the same local eligibility mechanism.

This factorization turns local credit assignment into a feedback-compatibility problem. Intrinsic low rank of the exact compartment-error matrix is useful, but it is not sufficient: the error field must also align with the particular feedback object available to the rule. We therefore distinguish global scalar broadcast, neuron-wise/per-soma broadcast, random low-rank feedback, path-structured feedback, and exact transported-error oracles. Dendritic branching creates the paths; E/I synapse banks set the local conductance state; and shunting inhibition changes the input resistance along each path. Shunting is not assumed to help monotonically. It helps restricted feedback when inhibitory conductance attenuates high-gain paths or otherwise makes the exact error field better aligned with the available broadcast.

Related work. This work builds on models of dendritic signal integration, branch-level nonlinear computation, divisive normalization, and biologically plausible teaching signals [34, 1, 3, 4, 35, 9, 8, 17, 18, 19, 26, 27, 28, 22, 45]. It also connects to machine-learning approaches that replace exact backpropagation with local, random, predictive, equilibrium, perturbation, or broadcast feedback signals [15, 14, 13, 16, 31, 21, 32, 33, 29, 30, 36, 37, 38, 39, 40]. Our contribution is complementary: we derive the exact conductance-tree credit object and test when shunting inhibition and dendritic morphology make that object broadcast-compatible.

We test this chain directly. Exact-gradient reconstruction verifies the factorization numerically. Gradient-fidelity diagnostics are used as diagnostics, not as the main claim of backprop equivalence: shunting has a higher final LocalCA to backprop cosine than the additive control, but the cosine remains modest, while scale mismatch is substantially reduced. Exact-error rank and direct broadcast-fidelity diagnostics show that shunting can make the compartment-error field easier for the implemented feedback object to approximate. Transported-error oracle

experiments show that supplying the exact effective path-gain-transported compartment error substantially reduces the local-learning gap, identifying feedback fidelity as an important bottleneck. In an MNIST gradient-dynamics diagnostic, final cosine is 0.100 ± 0.033 for shunting vs. 0.005 ± 0.026 for additive, and scale mismatch falls from 0.50 ± 0.05 to 0.26 ± 0.05 . On noise resilience under matched inhibition ($N_I \geq 5$), additive and shunting LocalCA are close under the practical per-soma broadcast, while supplying the [C] transported-error oracle lifts both substantially. This isolates feedback fidelity, rather than the eligibility rule alone, as the bottleneck.

Contributions. This paper makes three contributions: deriving the conductance-stage path gain, its local-eligibility/path-error factorization, and a closed-form inhibitory path-gain ratio; identifying path-gain and broadcast-compatibility metrics that mediate whether restricted somatic feedback can approximate exact compartment errors; and showing empirically that shunting inhibition can improve this approximation while also identifying regimes where scalar or low-rank feedback fails.

Claim status. We distinguish [A] analytical claims (factorization, path gains, inhibitory path-gain derivatives), [C] codebase/simulator diagnostics (exact reconstruction, broadcast comparisons, transported-oracle upper bounds, additive/shunting controls), and [B] biological predictions (branch-local inhibition and morphology-dependent credit routing). Transported-error results are always [C] oracle diagnostics, not proposed biological learning rules.

2 Compartmental Voltage Model and Gradient Derivation

For each dendritic compartment, we use a steady-state conductance model derived from discretized passive cable dynamics [1, 2]. In normalized units, leak and inhibitory reversal potentials are set to 0, the excitatory reversal potential is set to 1, and leak conductance is fixed at 1 (i.e., all conductances are expressed relative to leak). Each compartment computes a conductance-weighted voltage: excitatory conductance pulls the voltage toward the excitatory reversal potential, leak and shunting inhibition pull it toward zero, and dendritic coupling transfers child voltages upward toward the soma. This form makes local sensitivity depend on both driving force ($E - V$) and total input resistance R^{tot} . Throughout the paper we write dendritic morphologies as rooted trees $[b_1, b_2, \dots, b_D]$, where each factor gives the fan-in from one dendritic stage to the next on the path to the soma. Thus $[3, 3]$ means that each soma aggregates 3 proximal branches and each proximal branch aggregates 3 distal branches, for 9 distal leaves per soma. In the main feedforward experiments, synaptic inputs terminate on dendritic branches rather than directly on the soma. Each branch layer carries explicit excitatory conductance-based synapse banks and, when enabled, learned inhibitory branch-level conductance-based synapse banks driven by nonnegative inputs. This is input-driven shunting inhibition, not a recurrent lateral inhibitory circuit; explicit inhibitory-cell controls are reported in the appendix. The additive and shunting cores are architecture-matched at the tree level and differ only in the branch-level synaptic integration rule. These ingredients play distinct roles in the credit-assignment argument. Branching creates the path structure. E/I synapse banks determine the local conductance state. Shunting turns inhibitory conductance into a multiplicative change in path gain through input resistance. The local learning rule then asks whether a low-bandwidth teaching signal can stand in for the exact path-specific error induced by that tree. **Notation.** We use $\delta_{0,u}$ for the voltage-space error of soma u , δ_0 for the layer-wide soma/core error vector, $\bar{\delta}$ for the global scalar obtained by averaging soma coordinates within an example, and $\delta_{u,n} = \partial L / \partial V_{u,n}$ for compartment n of neuron u . We use α_n^{cond} for the conductance-stage path gain from compartment n to the soma and $\tilde{\alpha}_n$ for the effective path gain after including any post-voltage activation derivatives; when upward transfer is identity, $\tilde{\alpha}_n = \alpha_n^{\text{cond}}$. We use N_E and N_I for the number of excitatory and inhibitory synapses per branch, r_n^{4F} for the empirical 4F branch-soma covariance proxy, ϕ_n for the 5F confidence factor, λ_{mix} for the pathway-broadcast mixing weight, and κ_j for depth-dependent modulation in morphology extensions. All conductances and presynaptic activities in the main experimental setting are nonnegative; additive voltages may be signed comparators, but the conductance variables that define the shunting model are nonnegative.

Voltage equation and local sensitivities. Consider compartment n with synaptic inputs j (activity x_j , reversal E_j , conductance $g_j^{\text{syn}} \geq 0$) and dendritic inputs from children (voltage V_j , conductance $g_j^{\text{den}} \geq 0$). The steady-state voltage

is:

$$V_n = \frac{\sum_j E_j x_j g_j^{\text{syn}} + \sum_j V_j g_j^{\text{den}}}{\underbrace{\sum_j x_j g_j^{\text{syn}} + \sum_j g_j^{\text{den}} + 1}_{g_n^{\text{tot}}}}, \quad R_n^{\text{tot}} = 1/g_n^{\text{tot}}. \quad (1)$$

V_n is a convex combination of reversal potentials, child voltages, and leak. Writing $\mathcal{S}_n = \{0\} \cup \{E_j\}_j \cup \{V_j\}_j$, we have $\min \mathcal{S}_n \leq V_n \leq \max \mathcal{S}_n$ and $0 < R_n^{\text{tot}} \leq 1$. The local sensitivities follow directly:

Proposition 1 (Local Sensitivities).

$$\frac{\partial V_n}{\partial g_j^{\text{syn}}} = x_j R_n^{\text{tot}} (E_j - V_n), \quad \frac{\partial V_n}{\partial V_i} = g_i^{\text{den}} R_n^{\text{tot}}, \quad \frac{\partial V_n}{\partial g_i^{\text{den}}} = R_n^{\text{tot}} (V_i - V_n). \quad (2)$$

Eq. (2) gives the eligibility factors used by the local rules below: presynaptic activity or voltage difference, driving force, and input resistance. If a child transmits post-voltage activity $a_i = f_i(V_i)$ rather than raw voltage, the dendritic-conductance derivative is $R_n^{\text{tot}}(a_i - V_n)$ and the child-voltage error includes the local factor $f'_i(V_i)$.

Shunting inhibition as divisive gain control. An inhibitory synapse with $E_{\text{inh}} \approx 0$ contributes current $(0 - V_n)x_j g_j^{\text{syn}}$ and increases g_n^{tot} . Its sensitivity is $\partial V_n / \partial g_j^{\text{syn}} = -x_j R_n^{\text{tot}} V_n$, which corresponds to multiplicative attenuation (divisive normalization). Shunting is divisive at the voltage level, but its effect on firing rates can be subtractive in some regimes [10]. Inhibitory plasticity can balance excitation dynamically [11]; our learned inhibitory conductances serve an analogous role. The same denominator also gives the direct connection to path gain. Writing $G_n^E(x)$ and $G_n^I(x)$ for the total excitatory and inhibitory synaptic conductance impinging on compartment n ,

$$R_n^{\text{tot}}(x) = (1 + G_n^E(x) + G_n^I(x) + G_n^{\text{den}})^{-1}.$$

Increasing $G_n^I(x)$ lowers $R_n^{\text{tot}}(x)$ and therefore multiplicatively suppresses every upstream path whose error must pass through compartment n . In this sense inhibition can gate credit flow by changing the path gain, even when the inhibitory drive is feedforward rather than lateral. Prop. 2 formalizes this statement after the tree path gain is defined.

Exact gradients for dendritic trees. Let $V_{0,u}$ be the somatic output of neuron u and define its soma/core error as $\delta_{0,u} := \partial L / \partial V_{0,u}$. For a linear decoder $\hat{y} = W_{\text{dec}} V_0$, this reduces to $\delta_0 = W_{\text{dec}}^\top (\partial L / \partial \hat{y})$; for a nonlinear decoder, δ_0 is the corresponding decoder-input Jacobian product. The theoretical derivation is stated for the conductance-stage voltage before any post-voltage reactivation. When a branch applies a monotone reactivation before transmitting activity upward, the same recursion holds with additional local derivative factors along the path. All exact-error diagnostics below are computed as pre-activation voltage errors, including activation-derivative factors when reactivation is enabled.

Theorem 1 (Conductance-Stage Backpropagation on a Dendritic Tree). *For a rooted dendritic tree with soma at node 0 and unique parent $p(n)$ for each non-somatic compartment, suppose the parent receives either the pre-activation voltage V_n or a local post-voltage activity $a_n = f_n(V_n)$. For identity upward transfer, the loss gradient satisfies*

$$\frac{\partial L}{\partial V_n} = \frac{\partial L}{\partial V_{p(n)}} R_{p(n)}^{\text{tot}} g_{n \rightarrow p(n)}^{\text{den}}, \quad (3)$$

with boundary condition $\partial L / \partial V_0 = \delta_0$. With post-voltage transfer $a_n = f_n(V_n)$, the local factor becomes $f'_n(V_n) R_{p(n)}^{\text{tot}} g_{n \rightarrow p(n)}^{\text{den}}$. Because the graph is a tree, each compartment has a unique path to the soma. Defining the conductance-stage path gain

$$\alpha_n^{\text{cond}} = \prod_{(i \rightarrow k) \in \text{path}(n \rightsquigarrow 0)} R_k^{\text{tot}} g_{i \rightarrow k}^{\text{den}}, \quad \frac{\partial L}{\partial V_n} = \alpha_n^{\text{cond}} \delta_0, \quad (4)$$

with $\alpha_0^{\text{cond}} = 1$ under identity transfer. In the post-voltage case, the exact effective path gain is

$$\tilde{\alpha}_n = \prod_{(i \rightarrow k) \in \text{path}(n \rightsquigarrow 0)} f'_i(V_i) R_k^{\text{tot}} g_{i \rightarrow k}^{\text{den}}, \quad \frac{\partial L}{\partial V_n} = \tilde{\alpha}_n \delta_0, \quad (5)$$

where $f'_i(V_i)$ is omitted for identity transfer.

Proof. Apply the chain rule on the tree-structured computation graph using Prop. 1. $\square$

The one-step recursion in Eq. (3) and the path product in Eq. (4) separate local conductance transfer from the non-local somatic error for one soma/tree. Replacing $\delta_{0,u}$ by a layer-wide vector, a per-soma broadcast, or a single global scalar is an additional feedback-compression step introduced by LocalCA, not a consequence of the theorem itself.

Proposition 2 (Inhibitory control of conductance-stage path gain). *Let $\mathcal{A}(n)$ be the set of compartments on the path from compartment n to the soma, excluding n and including the parent compartments through which the error must pass. Holding dendritic coupling conductances fixed, an added inhibitory conductance $\Delta G_k^I(x) \geq 0$ at any $k \in \mathcal{A}(n)$ changes the conductance-stage path gain by*

$$\frac{\alpha_n^{\text{cond}}(x; \Delta G^I)}{\alpha_n^{\text{cond}}(x; 0)} = \prod_{k \in \mathcal{A}(n)} \frac{g_k^{\text{tot}}(x)}{g_k^{\text{tot}}(x) + \Delta G_k^I(x)}, \quad \frac{\partial \log \alpha_n^{\text{cond}}}{\partial G_k^I} = \begin{cases} -R_k^{\text{tot}}, & k \in \mathcal{A}(n), \\ 0, & k \notin \mathcal{A}(n). \end{cases} \quad (6)$$

Proof. Substitute $R_k^{\text{tot}} = 1/g_k^{\text{tot}}$ into the path-gain product in Eq. (4). Adding inhibitory conductance changes only the denominator at compartments on the path from n to the soma, which gives the product ratio and the log derivative in Eq. (6). $\square$

Prop. 2 makes two directed predictions. First, inhibition is a path gate: it attenuates all credit paths below the inhibited compartment. Second, inhibition is useful for restricted-feedback local learning only when this attenuation makes the distribution of α_n^{cond} more uniform or better aligned with the task. If inhibition is too strong, too heterogeneous, or generated by a poorly calibrated inhibitory population, it can suppress useful signals and hurt learning. In the additive control the same inhibitory input changes voltages and local eligibilities, but it does not enter the conductance-stage path recursion as a multiplicative resistance gate.

When does shunting compress path gains? The attenuation identity above is not by itself a concentration result. Let $z_n = \log \alpha_n^{\text{cond}}$ be the log conductance-stage path gain for compartment n . If added inhibitory conductance contributes a path-dependent attenuation

$$\beta_n(x) = \sum_{k \in \mathcal{A}(n)} \log \left(1 + \frac{\Delta G_k^I(x)}{g_k^{\text{tot}}(x)} \right), \quad z'_n = z_n - \beta_n, \quad (7)$$

then, over compartments or examples,

$$\text{Var}(z') = \text{Var}(z) + \text{Var}(\beta) - 2\text{Cov}(z, \beta). \quad (8)$$

Thus shunting narrows the log path-gain field exactly when

$$\text{Cov}(z, \beta) > \frac{1}{2} \text{Var}(\beta). \quad (9)$$

Eq. (7) follows by taking the logarithm of the product ratio in Prop. 2; Eq. (8) then applies the variance identity for $z - \beta$. Eq. (9) is a diagnostic condition under one averaging measure, not the empirical mechanism claimed below. In finite trained networks, the covariance margin can disagree with CV and exact-error-rank summaries because these statistics average over different compartment and sample axes. We therefore use direct path-gain dispersion, exact-error compressibility, and inhibition-intervention measurements as the empirical tests, and report the covariance margin as a boundary check in the appendix.

Corollary 1 (Local–Global Factorization). *The exact synaptic gradient at compartment n factorizes as:*

$$\frac{\partial L}{\partial g_i^{\text{syn}}} = \underbrace{x_i R_n^{\text{tot}} (E_i - V_n)}_{\text{synapse-local eligibility}} \cdot \underbrace{\frac{\partial L}{\partial V_n}}_{\text{compartment error}}, \quad (10)$$

where the eligibility term depends only on quantities available at synapse i (presynaptic activity x_i , input resistance R_n^{tot} , driving force $E_i - V_n$), and the compartment error $\partial L / \partial V_n$ is the sole non-local quantity.

The factorization in Eq. (10) isolates the fast task-error-dependent non-local quantity as $\partial L/\partial V_n = \alpha_n^{\text{cond}} \delta_0$ under identity upward transfer and $\partial L/\partial V_n = \tilde{\alpha}_n \delta_0$ when post-voltage transfer is enabled. The 4F and 5F factors introduced below use slower voltage statistics, so they are empirical reliability preconditioners rather than additional error signals. Any practical local rule therefore depends on how well its broadcast signal approximates that exact compartment error, which we evaluate in the gradient-fidelity diagnostic below. In the models studied here, shunting improves this approximation by reducing and stabilizing R_n^{tot} , which narrows the spread of local sensitivities across the tree. In the empirical networks, each dendritic branch layer may optionally apply a post-voltage reactivation $f(V_n)$ before passing activity to the next stage; unless otherwise noted, the manuscript sweeps use a learnable bounded `param_tanh` reactivation.

3 Local Learning Rules

Broadcast error approximation. We approximate the exact compartment error $\partial L/\partial V_n$ (Corollary 1) with a broadcast signal e_n derived from the somatic/core error δ_0 . For a linear readout this reduces to $\delta_0 = W_{\text{dec}}^\top (\partial L/\partial \hat{y})$; for a nonlinear decoder we use the corresponding decoder-input Jacobian product $J_{\text{dec}}(h_{\text{core}})^\top (\partial L/\partial \hat{y})$ so that the broadcast remains defined in soma/core coordinates. The main competence and mechanism runs use the implemented `per_soma` mode: when soma/core dimension matches a layer’s neuron dimension, each neuron receives its own $\delta_{0,u}$ and the same value is shared across that neuron’s dendritic compartments. A stricter scalar mode compresses the soma/core vector to one number per example, $\bar{\delta}_b = d_0^{-1} \sum_{c=1}^{d_0} \delta_{0,bc}$, and broadcasts that number to all neurons and compartments. These are different compression problems: per-soma broadcast tests within-tree path-gain approximation, whereas global scalar broadcast also discards heterogeneity across soma-error coordinates. For higher-bandwidth controls, P_K is a fixed random projection from soma-error coordinates to K feedback channels, $q_{n,k}$ are fixed compartment-specific mixing coefficients, and Γ_n denotes the pathway-structured transport map for compartment n . We distinguish five feedback objects: **global scalar broadcast**, $e_n = \bar{\delta} \mathbf{1}$; **neuron-wise/per-soma broadcast**, $e_{u,n} = \delta_{0,u}$ shared across the compartments of neuron u ; **rank- K random broadcast**, $c = P_K \delta_0$ and $e_n = \sum_k q_{n,k} c_k$; **path-structured broadcast**, $e_n = \lambda_{\text{mix}} \bar{\delta} \mathbf{1} + (1 - \lambda_{\text{mix}}) \Gamma_n(\delta_0)$ with pathway roles motivated by vectorized dendritic teaching signals [44]; and the **transported oracle**, $e_n^{\text{transport}} = \tilde{\alpha}_n \delta_0 = \partial L/\partial V_n$, an analysis upper bound because it supplies the effective path gain from Theorem 1. Operationally, path-transport runs with reactivation enabled transport activation-space errors through parent reactivation slopes and multiply by the local $f'_n(V_n)$ before conductance updates; with identity transfer this reduces to $\alpha_n^{\text{cond}} \delta_0$. Thus the main feedforward result tests neuron-wise/per-soma feedback rather than a single layer-wide scalar. Under shunting, this restricted field supports nontrivial learning on the standard classification tasks, while scalar, routed, and rank-bridge controls in the appendix expose regimes where lower-bandwidth feedback collapses branch identity and richer feedback is needed. A local-mismatch heuristic is included only as an appendix control because it performs much worse than the somatic-error broadcasts (Appendix A).

Three-factor learning rule (3F).

Definition 1 (3F Update). *For synaptic and dendritic conductances:*

$$\widehat{\nabla}_{g_j^{\text{syn}}} L = \langle x_j R_n^{\text{tot}} (E_j - V_n) e_n \rangle_B, \quad \widehat{\nabla}_{g_j^{\text{den}}} L = \langle R_n^{\text{tot}} (V_j - V_n) e_n \rangle_B, \quad (11)$$

where $\langle \cdot \rangle_B$ denotes the batch average.

The three factors are presynaptic activity x_j (or a voltage difference), postsynaptic modulation through driving force and input resistance, and the broadcast error e_n . The same rule applies to excitatory and inhibitory synapses. The sign difference comes from the driving force $(E_j - V_n)$. We write $\widehat{\nabla}_g L$ as the local gradient estimate supplied to the optimizer; the optimizer applies the usual learning rate and descent step. Equivalently, one could absorb the sign convention into e_n and treat it as a descent signal.

Additive control. Rule (11) is the local gradient of the *shunting* voltage $V_n = \sum E_j g_j x_j / g_n^{\text{tot}}$. For the additive E/I control, inhibition enters as a fixed signed voltage contribution rather than through the conductance denominator:

$$V_n^{\text{add}} = \sum_{j \in E} g_j^E x_j^E - \sum_{j \in I} g_j^I x_j^I + \sum_k g_k^{\text{den}} V_k.$$

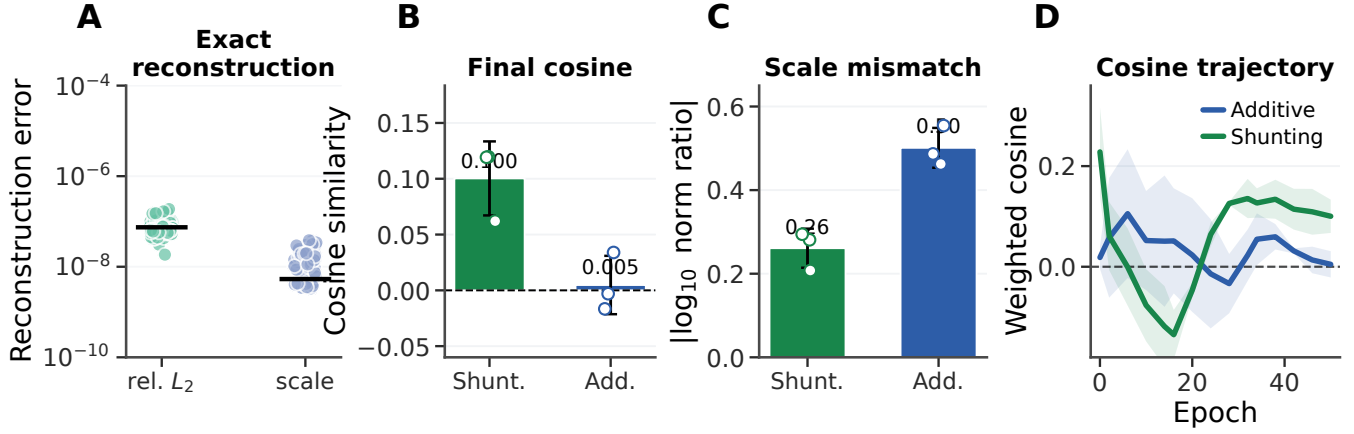


Figure 2: **Exact reconstruction and gradient diagnostics.** (A) Exact-factorization reconstruction matches autograd to numerical precision. (B) Final weighted LocalCA to backprop cosine, reported as a diagnostic rather than the main mechanism claim; dots show seeds. (C) Final scale mismatch, $|\log_{10}(\|g^{\text{local}}\|/\|g^{\text{bp}}\|)|$; dots show seeds. (D) Cosine trajectory over learning; Appendix Fig. S4 shows finite gradient norms. Error bars are ± 1 s.d. across 3 seeds.

Equivalently, $\partial V_n^{\text{add}}/\partial g_j^{\text{syn}} = s_j x_j$ with $s_j = +1$ for excitatory and $s_j = -1$ for inhibitory synapses, and $\partial V_n^{\text{add}}/\partial g_k^{\text{den}} = V_k$. The additive local gradient therefore has no driving-force or R_n^{tot} terms ($R_n^{\text{tot}} \equiv 1$ by definition since there is no denominator). Throughout, each architecture uses the learning rule derived from its own forward-pass dynamics. This keeps the comparison architecture-matched rather than applying a shunting-derived rule to an additive model. In implementations with unconstrained raw parameters θ and $g = \text{softplus}(\theta)$, Eqs. (11)–(12) give conductance-space gradients; before assigning gradients to θ , we multiply by $\partial g/\partial \theta$. The exact-gradient reconstruction diagnostics include this parameterization factor. The compact mechanism table below summarizes the matched shunting/additive comparison.

Property	Shunting / conductance core	Additive control
Voltage	Normalized by total conductance	Signed E/I voltage sum
Inhibition	Divisive conductance load	Subtractive signed term
Synaptic eligibility	$xR^{\text{tot}}(E - V)e$	sxe , $s \in \{+1, -1\}$
Driving force	Present through $E - V$	Absent from additive derivative
Input resistance	State-dependent, $R^{\text{tot}} = (g^{\text{tot}})^{-1}$	Effectively fixed
Credit interpretation	Inhibition changes path gain and voltage sensitivity	Inhibition shifts voltage but does not gate path gain

Practical heuristic wrappers: 4F and 5F. **4F (branch-soma relevance proxy).** In the conductance-stage recursion (4), the path gain α_n^{cond} attenuates the somatic error differently at each compartment. To compensate without explicitly computing that gain, 4F uses an online correlation proxy

$$r_n^{4F} = \text{Cov}(\bar{V}_n, \bar{V}_0) / (\sqrt{\text{Var}(\bar{V}_n) \text{Var}(\bar{V}_0)} + \epsilon).$$

Positive r_n^{4F} up-weights branches whose voltages covary with the soma; weak or sign-opposed covariance reduces the reliability of the shared broadcast for that branch. In the current implementation the online estimate is clamped to the positive stability range $[0.1, 2.0]$, so sign-opposed covariance reduces the multiplier rather than reversing the update direction. We keep 4F as a diagnostic intermediate rather than a theorem-derived path-gain estimate. Empirically, this factor alone gives little improvement over 3F; the practical jump comes from the 5F confidence factor below.

5F (confidence-weighted 4F). 5F adds a second heuristic factor

$$\phi_n = \text{Var}(V_n) / (\sigma_{\text{res}}^2 + \epsilon),$$

where σ_{res}^2 is the residual variance of V_n after linear regression on the parent voltage. Operationally, ϕ_n is a branch-wise confidence gate, clamped to $[0.25, 4.0]$ for stability. This factor is empirical, not theorem-derived; its gain should be read as practical denoising rather than theoretical necessity. It is positive and bounded, and rescales locally estimated

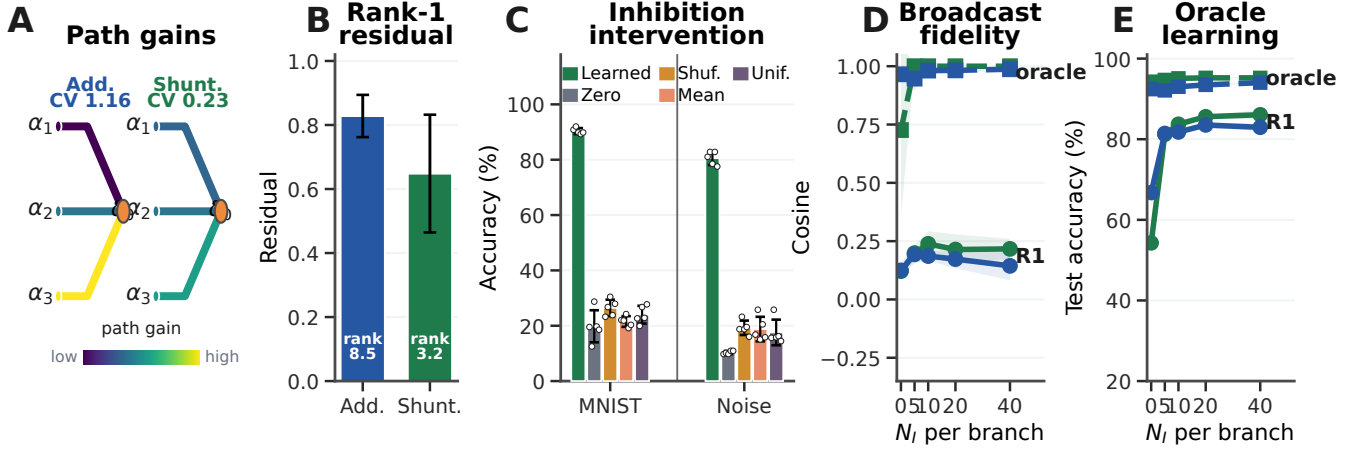


Figure 3: **Mechanistic chain from path gains to learning.** (A) Path-gain fields at matched inhibition. (B) Intrinsic exact-error compressibility on MNIST; this unrestricted SVD diagnostic is distinct from the fixed broadcast actually used. (C) Post-training inhibitory-conductance interventions (learned, zero, shuffled, mean-clamped, or uniform matched inhibition). (D) Compartment-error fidelity on noise resilience; dashed lines show the [C] transported-error oracle. (E) Transported-error learning where per-soma broadcast leaves a gap. Error bars are ± 1 s.d.

reliability from compartment and parent-voltage statistics rather than introducing a new error signal. Because the conditional-variance estimator can produce values above one, 5F is best interpreted as a bounded local preconditioner; the ablations in Fig. 5 and the sensitivity check in Appendix Fig. S2 are therefore essential parts of the evidence.

Definition 2 (5F Update).

$$\widehat{V}_{g_j^{\text{syn}}} L = r_n^{4F} \phi_n \langle x_j R_n^{\text{tot}} (E_j - V_n) e_n \rangle_B, \quad \widehat{V}_{g_j^{\text{den}}} L = r_n^{4F} \phi_n \langle R_n^{\text{tot}} (V_j - V_n) e_n \rangle_B. \quad (12)$$

Eq. (12) is therefore the main practical LocalCA update. The 3F rule uses the theorem-derived local eligibility with an approximate broadcast error, while 4F and 5F add empirical reliability factors. A feedback-alignment-style random-broadcast argument is included in Appendix C; it is supportive but not central to the main mechanism claim.

4 Experiments

Setup. The experiments test five linked predictions: exact factorization should reconstruct autograd gradients; shunting should improve the scale and structure of LocalCA credit signals even when directional cosine is weak; exact compartment errors should have lower intrinsic rank and better alignment with the implemented feedback object under shunting than under matched additive controls; learned inhibition should be functionally important and sample-dependent rather than merely adding a matched average conductance load; and replacing the broadcast with effective transported error should substantially reduce the local-learning gap. We evaluate capacity-calibrated supervised tasks and controlled sweeps that isolate inhibition strength, rule family, broadcast mode, morphology, and architecture. The main text uses MNIST [42], Fashion-MNIST [41], figure-ground MNIST (the context_gating configuration), noise resilience, and a compressed morphology-regime summary; CIFAR-10 [43] and routed-feedback diagnostics are reported in the appendix (Figs. S8, S12). Noise resilience is a nonnegative MNIST corruption task with a fixed latent Gaussian perturbation projected into input space and clipped to $[0, 1]$. Figure-ground MNIST places digit signal and distractor/context pixels in different halves of the input while preserving nonnegative drive; cue-routing in the appendix is the stronger context-dependent branch-credit task. Main comparisons are architecture-matched additive vs. shunting dendritic cores and local vs. backprop training; DFA is a supplementary baseline. Unless otherwise stated, local runs use 5F with per_soma feedback: each soma coordinate supplies the shared teaching signal for that neuron’s compartments. Global scalar, random low-rank, pathway-structured, and transported-error controls are labeled separately. All main-claim dendritic runs use nonnegative conductance transforms and nonnegative

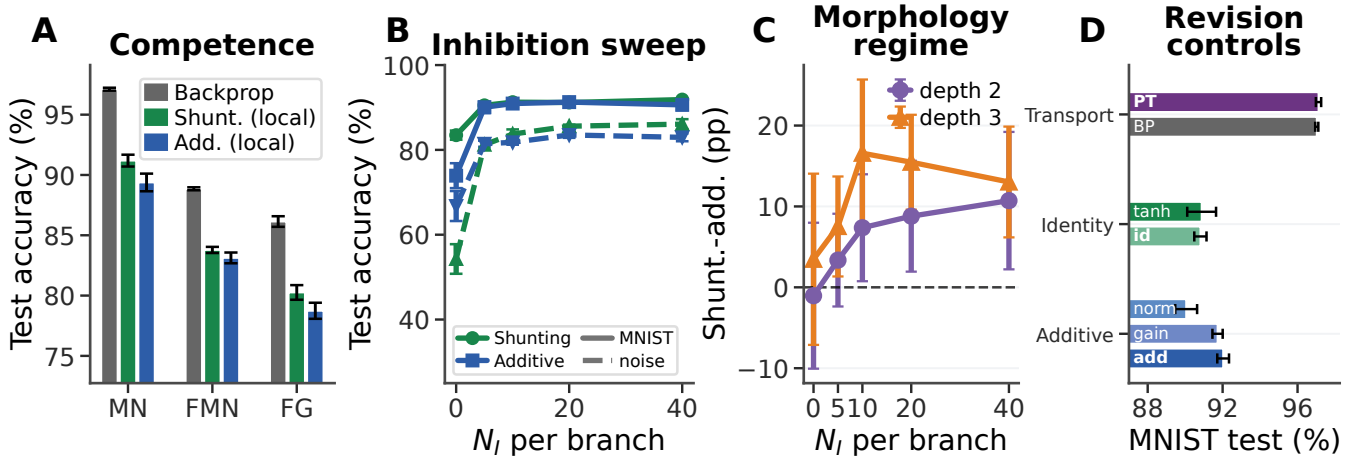


Figure 4: **Capacity-calibrated competence and regime dependence.** (A) Matched backprop vs. per-soma 5F LocalCA on MNIST, Fashion-MNIST, and figure-ground MNIST. (B) Inhibitory dose-response. (C) Morphology-dependent shunting advantage on noise resilience. (D) Revision controls on MNIST: exact path transport reaches the matched backprop ceiling, identity upward transfer preserves LocalCA competence, and additive gain/normalization variants do not exceed the additive baseline. Error bars denote ± 1 s.d. where available.

first-layer synaptic drive; image inputs are in $[0, 1]$, corrupted inputs are clipped back to $[0, 1]$, and figure-ground MNIST uses a nonnegative distractor-half construction. For figure-ground MNIST, the main LocalCA competence runs use an HSIC auxiliary term with weight 0.01; the appendix reports the HSIC ablation. The transported-error condition is an oracle upper bound derived from Theorem 1, not the proposed biological rule. Main mechanistic, oracle, and competence results use 5 seeds per condition unless noted otherwise; Appendix A defines the weighted gradient, path-gain CV, and exact-error SVD metrics.

From exact factorization to credit-signal fidelity. To test whether these performance differences correspond to better credit signals, we compare local and backprop gradients on the same batch at the same parameter values. For each parameter tensor p , we compute directional alignment (cosine similarity) and scale mismatch, defined as $|\log_{10}(\|g_p^{\text{local}}\|/\|g_p^{\text{bp}}\|)|$, aggregated by parameter count.

Across the MNIST gradient-dynamics diagnostic, shunting is closer to backprop than the additive control at the final checkpoint, but the directional agreement is still weak in absolute terms: weighted cosine rises from 0.005 ± 0.026 to 0.100 ± 0.033 , and scale mismatch falls from 0.50 ± 0.05 to 0.26 ± 0.05 . Figure 2A confirms that the implementation matches the theory under the nonnegative-input setup: reconstructing gradients from Corollary 1 using exact compartment errors yields weighted relative reconstruction error below 2×10^{-7} and weighted scale mismatch below 4×10^{-8} across runs. In a few low-norm parameter blocks, cosine becomes numerically unstable even though the reconstruction itself is essentially exact, so we report the norm-based diagnostics in the figure. Figure 2D shows the same gradient-alignment comparison over training: cosine is a noisy diagnostic and can be transiently weak, while additive alignment remains near zero at the final checkpoint rather than failing only at one static snapshot. Appendix Fig. S4 verifies that the corresponding local and backprop gradient norms remain finite throughout learning, ruling out a trivial zero-gradient explanation. At the level of the 3F rule, the central non-local approximation is therefore the broadcast substitute for $\partial L/\partial V_n$.

Mechanistic chain: path gains, broadcast fidelity, and oracle transport. Figure 3 provides the central empirical chain. At matched inhibition ($N_I=5$), shunting narrows the MNIST conductance-stage path-gain field (CV 0.23 vs. 1.16 for additive) and lowers the intrinsic rank of the exact compartment-error matrix: the unrestricted best rank-1 residual drops from 0.83 ± 0.07 to 0.65 ± 0.18 , while participation rank drops from 8.5 ± 3.3 to 3.2 ± 1.2 . This SVD result measures intrinsic geometry, not compatibility with a fixed scalar or per-soma broadcast field. We therefore pair it with the direct broadcast-fidelity and transported-oracle diagnostics in Fig. 3D,E, and the diagnostic pipeline reports shared-field and actual-broadcast residuals for the implemented feedback modes.

The inhibition intervention shows that the trained computation depends on the learned inhibitory field, rather

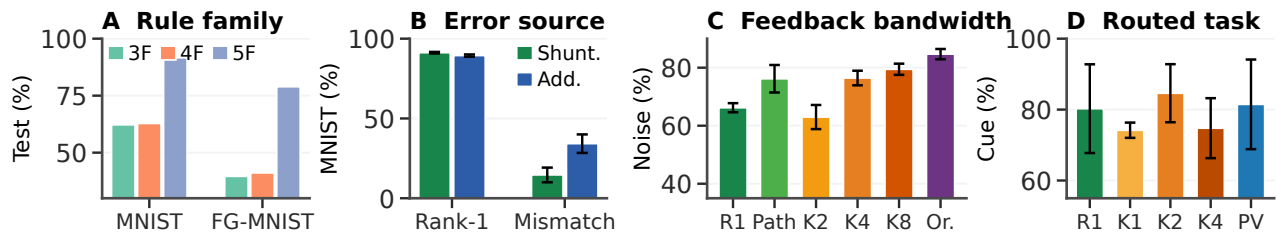


Figure 5: **Rule and feedback controls.** (A) Eligibility-rule ablation: 3F and 4F learn weakly, while the 5F reliability preconditioner is the practical LocalCA rule used in the competence results. (B) Error-source ablation: soma-derived feedback is essential, whereas local-mismatch feedback is not a usable substitute. (C) Feedback-bandwidth ladder on noise resilience: per-soma feedback, path-propagation, random low-rank feedback, and [C] exact transported error form a monotonic fidelity ladder. (D) Cue-routing feedback-rank diagnostic: richer feedback helps relative to the restricted baseline, while pathway-vector feedback remains an exploratory structured control.

than only on a matched average conductance load: zeroing, shuffling, or clamping learned branch inhibition drops MNIST from about 90% to 20–26% and noise resilience from about 81% to 10–19%. The fidelity and oracle panels probe the same broadcast bottleneck: per-soma feedback tracks the exact compartment error better under shunting once inhibition is present, transported error reaches about 0.95–1.0 fidelity, and transported learning lifts both the shunting and additive networks to 92–95% on noise resilience. CIFAR-10 and direct-I versus explicit-I path-gain probes remain appendix diagnostics (Fig. S8; Table S7).

Capacity-calibrated competence. In the capacity-calibrated regime, shunting LocalCA remains within 5–6 percentage points of matched backpropagation on MNIST, Fashion-MNIST, and figure-ground MNIST (Table S5; Fig. 4). The revision controls in Fig. 4D sharpen this interpretation: exact path transport reaches the matched shunting backprop ceiling on MNIST across 3F/5F and decoder-update choices, the identity-transfer version of the shunting core matches the post-voltage param_tanh version, and additive gain/normalization variants do not outperform the additive 5F per-soma baseline. Thus the central bottleneck is the fidelity of the feedback field to the exact compartment error, not a generic additive gain or voltage-normalization effect. Fig. 5 summarizes the broader rule and feedback controls: 5F is the strongest practical optimizer, soma-derived feedback is essential, and the noise-resilience rank bridge improves as feedback bandwidth increases from low-rank $K=1$ to $K=8$, with [C] path transport highest.

Regime dependence of the shunting advantage. With matched per-soma feedback, shunting is modestly better on MNIST, Fashion-MNIST, and figure-ground MNIST, including 0.803 ± 0.006 vs. 0.787 ± 0.007 for additive on figure-ground MNIST. The gap grows on inhibition-sensitive noise resilience once inhibitory conductance is present (Fig. 4B). Morphology shifts the useful inhibitory operating regime: different tree depths and branch factors peak at different N_I , and the shunting advantage is not monotonic in inhibition (Fig. 4C). Appendix Fig. S6 extends this boundary check to depth scaling and noisy error broadcasts.

5 Discussion and Limitations

We studied local credit assignment in conductance-based dendritic networks with explicit E/I conductance synapses, shunting inhibition, and tree-structured branching. Exact gradients separate synapse-local eligibility–presynaptic activity, driving force, and input resistance—from one path-specific compartment error, making LocalCA an error-approximation problem. The experiments support a coherent mechanism: branching creates credit paths, E/I synapse banks set the conductance state, and shunting modulates input resistance along those paths. In the regimes studied here this lowers intrinsic exact-error rank, improves direct feedback-fidelity diagnostics, reduces gradient-scale mismatch, and lets transported-error oracles substantially reduce the local-learning gap. Post-training inhibition interventions further show that learned inhibitory conductance is sample-dependent and functionally important to the trained forward computation.

The operating regime matters. Shunting is not universally beneficial, more inhibition is not always better, and the practical 5F rule is an empirical bounded preconditioner. The core theorem covers conductance-stage branch

parameters; reactivation, decoder, router, top- k , and HSIC terms are documented as implementation-specific updates. Eq. (9) is interpretive, and activation shape, morphology, task structure, and feedback rank can all change broadcast compatibility. We therefore claim a credit-geometry mechanism supported by exact reconstruction, fidelity diagnostics, controls, and oracle transport, not competitive CIFAR-10 performance, recurrent inhibitory circuits, or sufficiency of a single global scalar broadcast for routed/pathway-specific tasks.

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Diagnostic	Regime	Cached source	Use in paper
Gradient / N_I sweep	Nonnegative inputs, [128, 128] E layers, [3, 3] tree, 5F per-soma feedback, 5 seeds	analysis/gradient_fidelity_vs_ie_nonnegativeinput_fix/gradient_fidelity_summary.csv	Active Fig. 4B and supplementary N_I details
Theory diagnostics	Same nonnegative-input sweep, hooks for exact errors and path gains	analysis/theory_diag_gradient_fidelity_vs_ie_nonnegativeinput_fix_summary/theory_diag_by_condition.csv	Active path-gain, fidelity, and exact-reconstruction diagnostics
Rank bridge	Noise resilience, shunting, low-rank $K=1, 2, 4, 8$ and path transport, 5 seeds	analysis/rank_bridge_nonnegativeinput_fix/noise_resilience_rank_bridge_summary.csv	Fig. 5C and Table S9
Broadcast-noise stress	Separate error-noise stress diagnostic	analysis/noise_robustness.csv	Appendix stress-test boundary only
Older corrected summary	Different analysis provenance	data/gradient_fidelity_vs_ie_corrected_summary.csv	Not used for current claims

Table S1: **Noise and gradient diagnostic provenance.** The active manuscript claims use the nonnegative-input refreshes; older corrected summaries are retained in the repository for audit history but are not mixed with the current figures.

A Supplementary Results and Diagnostics

This appendix records the supporting diagnostics, implementation details, and task protocols needed to interpret and reproduce the paper’s main claims.

Appendix structure. The appendix is organized to make the paper reproducible from mechanism to implementation. Appendix A first reports the supplementary diagnostics that support or delimit the main empirical claims. Appendix B then specifies the implemented dendritic networks, including morphology, synapse banks, shunting and additive forward equations, positive parameterization, reactivation, decoder errors, and optimizer-level gradient assignment. Appendix C collects learning-rule variants and auxiliary theory details that are useful for reproducing the local updates but are not needed for the main proof. Appendix D gives task construction and robustness protocols.

Boundary diagnostics. Several controls are intentionally negative or mixed. The log-gain covariance margin from Eq. (9) is not used as a positive main result, and the pathway-vector feedback variant does not beat the best unstructured low-rank cue-routing control. These diagnostics delimit the claim rather than smoothing over failure modes.

Result Provenance and Roadmap

The main text uses only the nonnegative-input diagnostic refreshes for the gradient, noise-resilience, and path-transport claims. An older summary, data/gradient_fidelity_vs_ie_corrected_summary.csv, belongs to a different analysis provenance and is not used for the current main claims; in particular, its large noise-resilience additive–shunting gap should not be compared to the active nonnegative-input summaries. Tables S1–S3 make the data lineage explicit.

Diagnostic Metrics

The gradient-fidelity diagnostics compare LocalCA and backpropagation gradients at matched parameters and batches. For a parameter block p , let g_p^{local} and g_p^{bp} be the two gradients and let $w_p = \text{numel}(p) / \sum_q \text{numel}(q)$. The reported weighted cosine is

$$C_w = \sum_p w_p \frac{\langle g_p^{\text{local}}, g_p^{\text{bp}} \rangle}{\|g_p^{\text{local}}\|_2 \|g_p^{\text{bp}}\|_2 + \epsilon},$$

Claim / diagnostic	Script anchor	Cached source	Figure / table
Exact factorization reconstructs autograd	scripts/measure_theory_diagnostics.py; scripts/generate_neurips_figures.py	analysis/theory_diag_gradient_fidelity_vs_ie_nonnegativeinput_fix_summary/theory_diag_by_condition.csv	Fig. 2A; supplementary gradient figures
Path-gain and exact-error compressibility	scripts/generate_theory_diagnostics_figures.py; scripts/measure_error_rank_diagnostics.py	analysis/error_rank_selected_20260427/error_rank_summary.csv	Fig. 3A–B; Fig. S11
Inhibitory-conductance intervention	scripts/measure_inhibition_causality.py; scripts/generate_theory_diagnostics_figures.py	analysis/inhibition_causality_selected_20260427/inhibition_causality_runs.csv	Fig. 3C; Fig. S11
Transported-error oracle	scripts/summarize_path_transport_sweep.py; scripts/generate_theory_diagnostics_figures.py	analysis/path_transport_upper_bound_nonnegativeinput_fix_5seed/path_transport_upper_bound_summary.csv	Fig. 3D–E
Rule and feedback controls	scripts/generate_neurips_figures.py	analysis/rank_bridge_nonnegativeinput_fix/noise_resilience_rank_bridge_summary.csv and rule-control summaries	Fig. 5; Table S9
Capacity-calibrated competence	scripts/summarize_neurips_claim_sweeps.py; scripts/generate_neurips_figures.py	figures/data/competence_summary_20260422.csv; Fashion-MNIST competence summaries	Fig. 4; Table S5

Table S2: **Claim provenance.** Script anchors identify the analysis entry points or figure generators used to produce each cached diagnostic.

and weighted scale mismatch is

$$S_w = \sum_p w_p \left| \log_{10} \frac{\|g_p^{\text{local}}\|_2 + \varepsilon}{\|g_p^{\text{bp}}\|_2 + \varepsilon} \right|.$$

For exact-error compressibility, we form a matrix E whose rows are examples and whose columns are compartment–neuron coordinates of the exact pre-reactivation compartment error. If $E = U\Sigma V^\top$ and σ_i are the singular values, the unrestricted rank-1 residual and participation rank are

$$\rho_1 = \frac{\sqrt{\sum_{i>1} \sigma_i^2}}{\sqrt{\sum_i \sigma_i^2}}, \quad r_{\text{part}} = \frac{(\sum_i \sigma_i^2)^2}{\sum_i \sigma_i^4}.$$

This is an intrinsic geometry metric: both the sample coefficients and spatial template are chosen in hindsight by SVD. The constrained shared-field residual instead fixes the spatial template to all ones and allows only the best per-example scalar,

$$\rho_1 = \frac{\|E - \text{rowmean}(E)\mathbf{1}^\top\|_F}{\|E\|_F}.$$

The broadcast-conditioned residual uses the actual feedback field B generated by the implemented mode,

$$\rho_{\text{bc}} = \frac{\|E - B\|_F}{\|E\|_F},$$

with a corresponding flattened cosine. These constrained metrics are emitted by `scripts/measure_error_rank_diagnostics.py` alongside the unrestricted SVD metrics, so intrinsic error geometry and compatibility with the implemented broadcast field can be reported separately. Path-gain dispersion is the coefficient of variation of α_n^{cond} over the measured sample and compartment field. Compartment-error fidelity is the cosine between the tested broadcast field and the exact compartment-error field after flattening over the same sample and compartment axes.

Figure / control	Primary message	Supporting appendix material
Fig. 1	Model, factorization, and claim-type tags	Implementation equations; Table S10; Algorithm 1
Fig. 2	Exact reconstruction is precise; cosine is a weak diagnostic; scale mismatch improves	Fig. S4; Table S1; gradient metric definitions
Fig. 3	Path gains, exact-error rank, inhibition intervention, and transported-oracle chain	Fig. S11; Table S8; CIFAR and input-mode diagnostics
Fig. 4	Capacity-calibrated performance, regime dependence, and revision controls	Table S5; morphology map; additive-normalization and stress-test figures
Fig. 5	5F is practical, soma-derived feedback matters, and more feedback bandwidth helps	Table S4; Table S9; cue-routing diagnostics

Table S3: **Main-to-supplement roadmap.** The appendix is organized around the evidence needed to interpret each main figure rather than as a development log.

Rule and Feedback Controls

These controls separate three questions that are deliberately collapsed in the main competence summary: which local-rule wrapper is practical, whether the somatic error source matters, and how performance changes as the feedback field gains bandwidth or path information.

Rule-Family Ranking

This table keeps the 3F/4F/5F optimizer distinction explicit: the theorem-derived eligibility is necessary for the story, but the reported LocalCA competence results use the empirical 5F stabilizer.

Dataset	Rule	Mean valid, 10 validation-selected configs	Mean test, same 10 configs
MNIST	3F	0.611	0.622
MNIST	4F	0.620	0.628
MNIST	5F	0.912	0.916
Figure-ground MNIST	3F	0.398	0.396
Figure-ground MNIST	4F	0.411	0.411
Figure-ground MNIST	5F	0.807	0.789

Table S4: **Rule-family ranking.** Mean over the ten best validation-selected configurations from local-competence sweeps; this is not top-10 classification accuracy, and test scores are reported only after validation selection.

5F Stabilizer Sensitivity

Because the headline LocalCA runs use the empirical 5F stabilizer, we also tested whether performance depends sharply on the confidence clamp or the EMA rate used to estimate branch statistics. Across three-seed MNIST sensitivity checks, tightening or widening the clamp and varying the EMA rate leaves test accuracy near the default setting. This does not make 5F theorem-derived; it supports interpreting it as a bounded local reliability preconditioner rather than a brittle tuned error source.

Local Competence and Regime Dependence

Table S5 reports the capacity-calibrated headline competence numbers used in the main text. The point of this table is not to claim state-of-the-art benchmark performance, but to show the remaining LocalCA-to-backprop gap under matched dendritic capacity and per-soma 5F feedback.

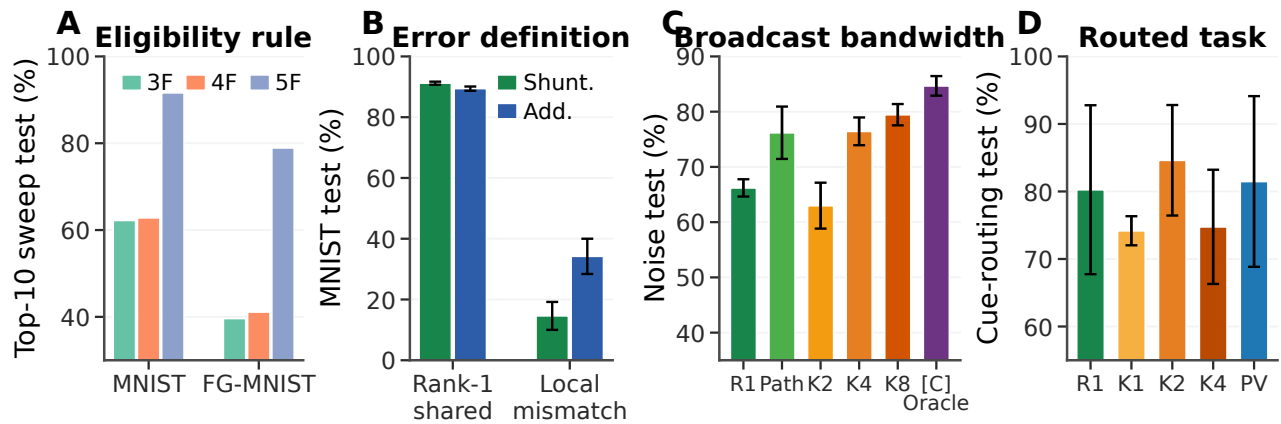


Figure S1: **Detailed rule and feedback controls.** (A) Eligibility-rule ablation: 3F, 4F, and 5F. (B) Error-source ablation: soma-derived feedback vs. local mismatch. (C) Feedback-bandwidth ladder on noise resilience: per-soma, recursive Path, K2/K4/K8, and [C] transported Oracle. (D) Cue routing: scalar/per-soma, random low-rank K1/K2/K4, and pathway-vector feedback.

Extended Gradient Analysis and N_I Sweep Detail

Figure 2 in the main text summarizes the final-state gradient diagnostic. The supplementary view below separates static scale checks, inhibitory dose-response curves, and individual Fashion-MNIST seed behavior so that the aggregate claims are not driven by a single summary panel.

Alignment dynamics are not a zero-gradient artifact. To check whether low additive alignment could be caused by vanishing gradients, we analyzed a separate 3-seed checkpoint trajectory with parameter-level LocalCA and backprop gradients. The additive local gradients are nonzero throughout training (weighted norm range 9.5×10^{-4} – 9.5×10^{-3}), and the corresponding backprop gradients are also nonzero (2.3×10^{-5} – 5.7×10^{-3}). The low additive cosine therefore reflects directional and scale mismatch, not absent gradients: additive 5F cosine stays near zero from epoch 0 to epoch 50 (0.018 to 0.005), while its local/backprop norm ratio falls from about 8.8×10^2 to 0.69.

Verification and Reproducibility

These checks compare the main seed set with held-out seeds and report the HSIC-weight sensitivity for figure-ground MNIST. The revision runs use the same objective-weight comparison on both seed sets, so seed robustness and HSIC contribution can be summarized as a matched two-factor control.

Additional Stress Tests

These stress tests probe two ways in which low-bandwidth broadcast can become unreliable: deeper dendritic paths and noisy teaching signals. They are not separate headline benchmarks; they delimit where shunting remains useful and where additive local learning degrades.

FA/DFA Baseline Comparison

The feedback-alignment baseline is included as an adjacent local-learning reference point. DFA can be evaluated on the dendritic cores, whereas the FA condition is marked unavailable for the block-structured dendritic updates because the required random feedback matrices are not dimensionally compatible with the branch-level parameterization.

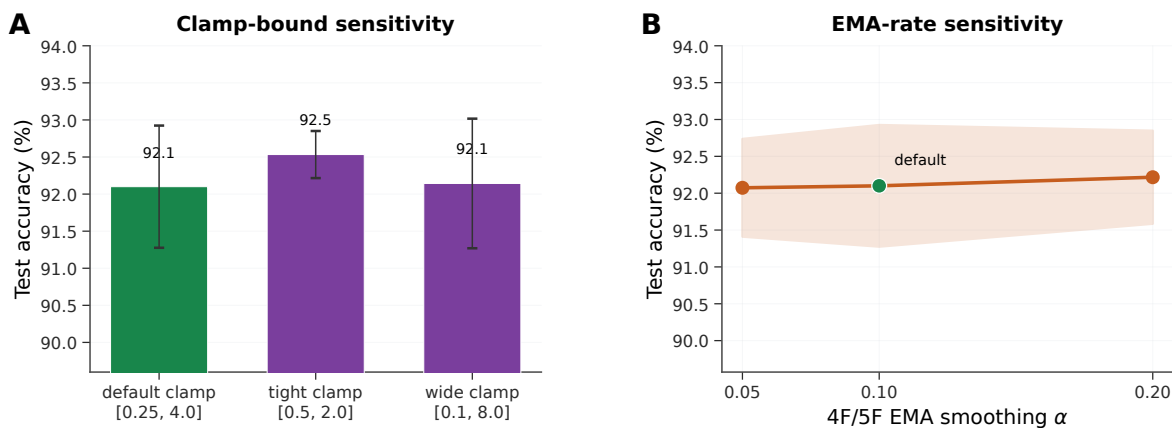


Figure S2: **5F stabilizer sensitivity.** (A) Changing the confidence-factor clamp from the default [0.25, 4.0] to tighter [0.5, 2.0] or wider [0.1, 8.0] bounds leaves MNIST test accuracy near 92%. (B) Changing the 4F/5F EMA smoothing rate over $\alpha \in \{0.05, 0.10, 0.20\}$ gives similarly stable performance. Error bars and bands are ± 1 s.d. across 3 seeds.

CIFAR-10 Results

Table S6 provides the exact values for the appendix CIFAR-10 figure. CIFAR-10 is used only as a harder-data mechanistic diagnostic: under flattened inputs, nonnegative conductance constraints, and a compact passive dendritic core, the matched backprop ceiling is the relevant reference rather than modern vision accuracy. In this stronger compact direct-I-stream family, shunting performs well under standard training (49.5% vs. 48.3% for additive), so the harder-data stress test does not unfairly penalize the shunting architecture. The broadcast-fidelity ladder also survives once decoder-aware soma mapping is used: additive LocalCA reaches 25.5% under per-soma feedback and 32.4% under transported error, while shunting rises from 29.7% under per-soma feedback to 35.0% with random low-rank $K=4$ and 47.2% with transported error, nearly matching the 49.5% shunting backprop ceiling. Removing learned I-to-E conductance lowers the shunting ceiling by 5.7 pp and transported LocalCA by 8.6 pp, making this a harder-data diagnostic for inhibitory conductance as part of the useful credit geometry rather than just an added architectural detail.

Additive Normalization Control

This control separates generic additive voltage normalization from shunting conductance dynamics in a lower-capacity MNIST stress regime.

Morphology $\times$ Inhibition Regime Map

This regime map tests whether tree geometry changes the inhibitory operating point at which shunting helps. It supports the claim that morphology changes the useful inhibition regime; a fuller mechanistic account would require matched theory diagnostics for every cell in the grid.

Input-Mode Probe: Direct Inhibitory Stream vs. Explicit Inhibitory Cells

The main experiments use input-driven inhibitory streams: the transfer layer provides nonnegative excitatory and inhibitory channels from the external input, and excitatory dendrites receive learned I-to-E conductance banks directly. To test whether the same mechanism can be generated by an explicit feedforward inhibitory population, we ran a one-feedforward-layer noise-resilience probe that preserves the relevant dendritic depth ([3, 3] morphology). In this setting, explicit inhibitory cells reach $94.2 \pm 0.2\%$ under path-transport LocalCA when their dendrites use the same update policy as excitatory dendrites, $93.8 \pm 0.2\%$ when those inhibitory-cell updates are frozen, and $95.1 \pm 0.3\%$

Dataset	BP ceiling	Shunting local	Additive local	Shunting BP gap (pp)
MNIST	0.971	0.912 ± 0.005	0.894 ± 0.007	5.9
Fashion-MNIST	0.889	0.838 ± 0.003	0.831 ± 0.004	5.1
Figure-ground MNIST	0.861	0.803 ± 0.006	0.787 ± 0.007	5.8

Table S5: **Local competence.** Backprop ceilings are matched-capacity shunting references from dedicated 5-seed standard-training refreshes for MNIST and figure-ground MNIST and from the dedicated 5-seed Fashion-MNIST competence sweep. Local values are 5F with per-soma feedback and local decoder updates. Figure-ground MNIST additionally uses an HSIC auxiliary objective (weight 0.01; see the HSIC auxiliary-objective appendix paragraph). Error bars on local values: ± 1 s.d. across 5 seeds; gaps are percentage points.

under standard backprop. Direct input-driven inhibition gives the expected broadcast-fidelity ladder on the same one-layer architecture: per-soma feedback reaches $85.2 \pm 0.7\%$, while exact effective path transport reaches $95.2 \pm 0.0\%$. Thus explicit inhibitory cells can generate useful path-gain structure once the architecture isolates the dendritic-path question.

Post-training inhibition intervention and exact-error rank. We next tested whether learned inhibitory conductance carries sample-dependent structure rather than only changing global scale. We replayed trained shunting LocalCA checkpoints while replacing the branch-level inhibitory conductance with four post-training interventions: zero inhibition, sample-shuffled inhibition, per-branch batch means, or one uniform matched mean. On a 512-example held-out subset, learned MNIST inhibition at $N_I=5$ gives $90.4 \pm 1.1\%$ accuracy, while the four interventions drop to 19.8–26.4%. On noise resilience at $N_I=10$, learned inhibition gives $80.5 \pm 2.4\%$, while the interventions drop to 10.3–19.3%. Thus the inhibitory field is not interchangeable with a matched global conductance load. We also computed the SVD of exact compartment-error matrices across samples and compartments. At matched MNIST $N_I=5$, shunting has lower unrestricted rank-1 residual than additive (0.65 ± 0.18 vs. 0.83 ± 0.07) and lower participation rank (3.2 ± 1.2 vs. 8.5 ± 3.3), supporting but not by itself proving the path-gain compressibility interpretation.

We also computed the direct log-gain covariance margin in Eq. (9) on the selected morphology diagnostic checkpoints. The margin is zero when no inhibitory conductance is present and slightly negative in the inhibited shunting selections (-2.5×10^{-3} , -5.0×10^{-4} , and -2.8×10^{-4} at $N_I=5, 20, 40$). We therefore do not use this covariance condition as a positive main result. Its role is to clarify when shunting should compress gains; the empirical claim rests on the measured comparative path-gain dispersion, exact-error rank, and inhibition-intervention diagnostics.

Experimental Predictions

The model is not a detailed cell-type circuit model, but its path-gain identity gives falsifiable dendritic predictions. These predictions are naturally aligned with branch-level dendritic computation and compartment-targeted inhibition in pyramidal neurons [5, 6, 7, 12].

Cue-Routing Feedback-Rank Diagnostic

Cue routing tests a regime where one shared teaching signal is expected to fail: context determines which noisy cue is reliable, so different pathways should receive different credit. Figure S12 shows that the benchmark is learnable and that moving beyond the restricted per-soma / single-channel baselines helps, but the path-structured variant does not yet beat the best unstructured low-rank setting. We therefore interpret this result as a failure-mode and rank-requirement diagnostic, not as a solved structured-feedback method.

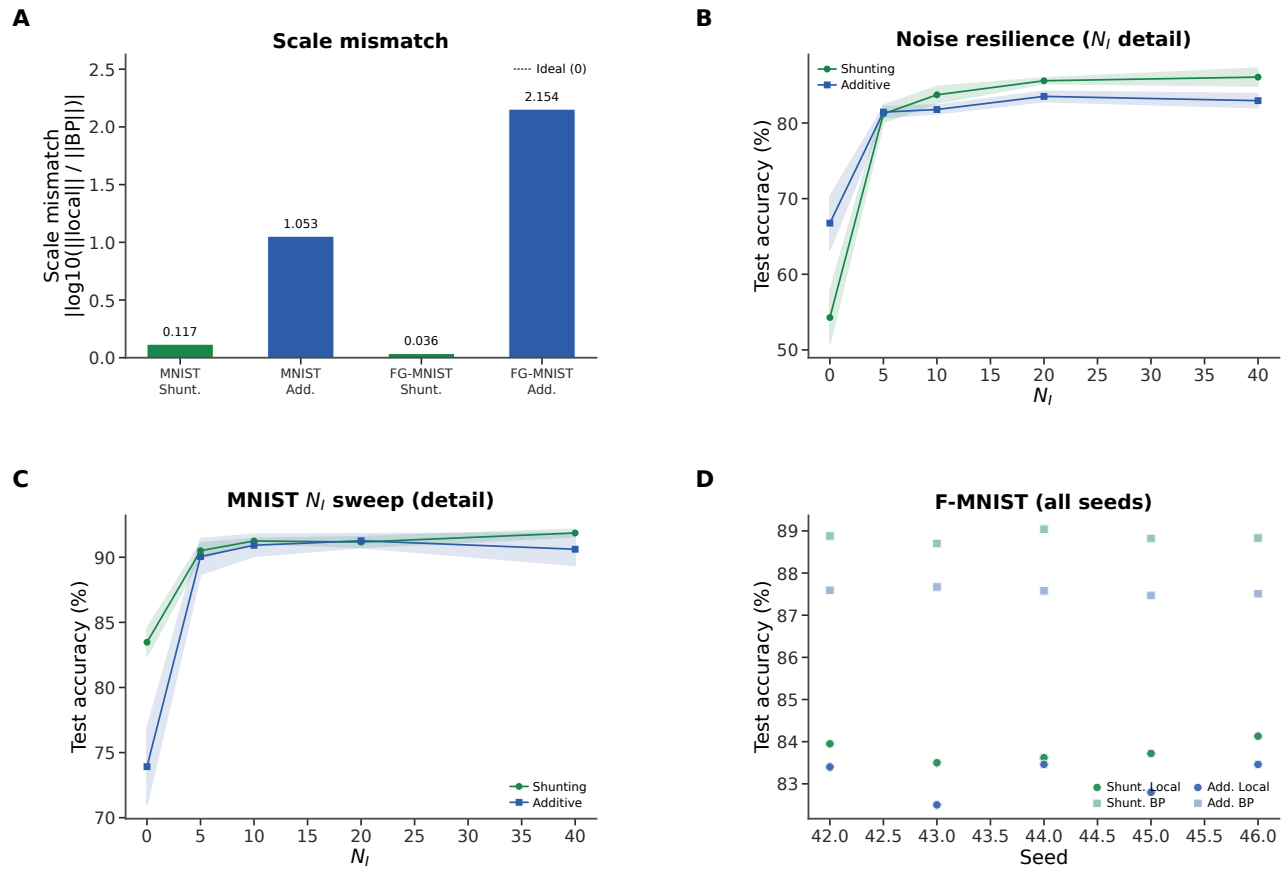


Figure S3: **Extended gradient and N_I analysis (supplementary).** (A) Log-scale mismatch, $|\log_{10}(\|g^{\text{local}}\|/\|g^{\text{bp}}\|)|$, from a separate matched-weight static diagnostic; these values should not be compared directly to the checkpointed final-state diagnostic in Fig. 2. The ideal value is 0; shunting is near ideal (0.117), whereas additive shows order-of-magnitude distortion (> 1.0). (B) Noise-resilience N_I dose-response with error bands (± 1 s.d.). (C) MNIST N_I dose-response detail with error bands. (D) Fashion-MNIST individual seeds for all conditions, showing consistency across runs.

Random Low-Rank Broadcast Channels

These controls test whether failures of the restricted per-soma baseline reflect insufficient feedback bandwidth rather than a failure of local eligibility itself.

B Dendritic Network Implementation

This section gives the implementation-level specification needed to reproduce the dendritic architectures and LocalCA updates. It separates the biological modeling assumptions from software choices such as positive raw-parameter transforms, decoder-error coordinates, and optimizer gradient assignment.

Implementation Overview

Each feedforward dendritic layer is a population of dendritic neurons. A neuron contains a rooted tree of compartments; each compartment has an excitatory synapse bank, optionally an inhibitory synapse bank, and dendritic coupling conductances from its children. The main experiments use branch-level input-driven inhibition: a nonnegative transfer stage produces excitatory and inhibitory input streams from the external input, and the inhibitory stream drives learned I-to-E conductance banks on excitatory dendritic branches. The explicit-I probe in Table S7 instead instantiates a

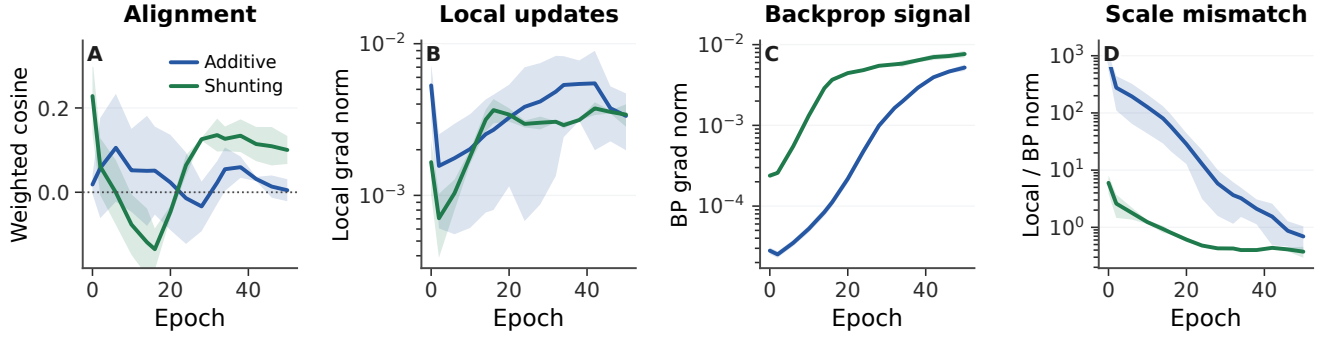


Figure S4: **Alignment dynamics with explicit gradient-norm checks.** (A) Weighted LocalCA to backprop cosine over training for the 5F rule. (B,C) Weighted local and backprop gradient norms stay nonzero for both additive and shunting models. (D) The additive control begins with a much larger scale mismatch, then approaches the backprop scale without recovering strong directional alignment. Shading is s.d. across 3 seeds.

separate feedforward inhibitory population, whose output projects inhibitory conductance onto excitatory dendrites.

The additive and shunting cores are architecture-matched: they use the same tree, synapse counts, nonnegative transfer stage, positive conductance parameterization, optimizer protocol, and seed sets. They differ only in how a branch combines excitatory, inhibitory, and dendritic inputs. Decoder weights are unconstrained readout parameters and are not interpreted as conductances.

Morphology and Synapse Banks

The morphology notation $[b_1, b_2, \dots, b_D]$ specifies a rooted dendritic tree. The factor b_d is the fan-in from dendritic stage d to the next more proximal stage. Thus $[3, 3]$ gives 3 proximal branches per soma and 3 distal branches per proximal branch, for 9 distal leaves per soma. In the main feedforward models, synaptic inputs terminate on dendritic branches rather than directly on the soma. Unless a control explicitly changes this, each branch has N_E excitatory synapses and N_I inhibitory synapses. The branch input is sparse through a learned top- k or banked synapse selection module; inactive synapses are masked out in the local update unless the corresponding control enables inactive-weight updates.

For a branch n , write $x_{nj}^E, x_{nj}^I \geq 0$ for the selected excitatory and inhibitory presynaptic drives and a_c for the post-reactivation activity of child compartment c . The branch-level excitatory, inhibitory, and dendritic conductance currents are

$$U_n^E = \sum_{j=1}^{N_E} g_{nj}^E x_{nj}^E, \quad U_n^I = \sum_{j=1}^{N_I} g_{nj}^I x_{nj}^I, \quad D_n = \sum_{c \in \text{child}(n)} g_{c \rightarrow n}^{\text{den}} a_c.$$

The corresponding dendritic coupling load is

$$G_n^{\text{den}} = \sum_{c \in \text{child}(n)} g_{c \rightarrow n}^{\text{den}}.$$

For leaf branches, $D_n = G_n^{\text{den}} = 0$. For direct input-driven inhibition, x^I is produced by the inhibitory transfer stream. For explicit-I controls, x^I is the activity of the inhibitory-cell population.

Shunting and Additive Forward Equations

In the shunting core, excitatory reversal is 1, leak and inhibitory reversal are 0, and leak conductance is fixed to 1. The implemented branch voltage is therefore

$$V_n^{\text{shunt}} = \frac{U_n^E + D_n}{1 + U_n^E + U_n^I + G_n^{\text{den}} + \epsilon}.$$

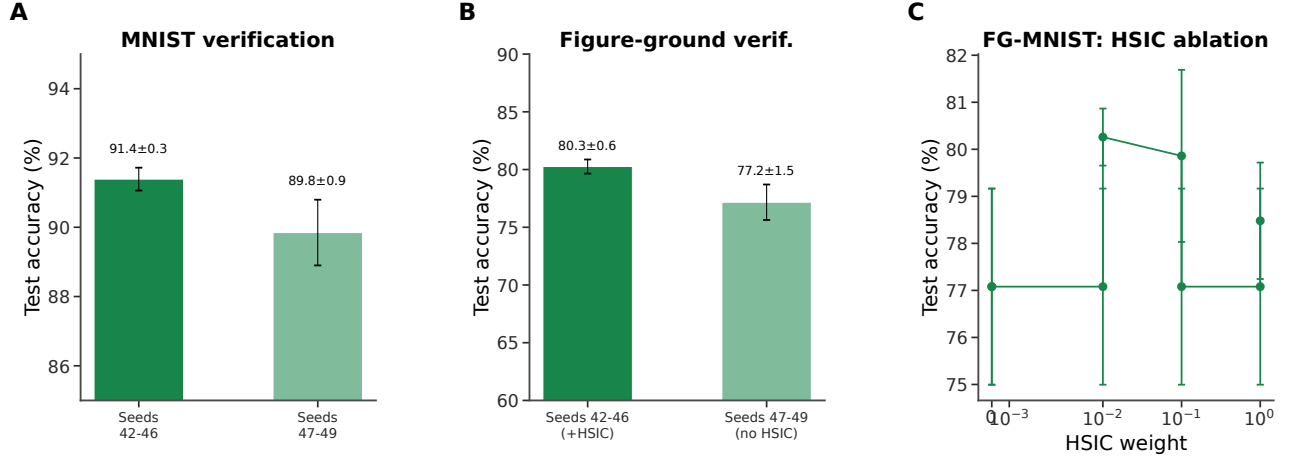


Figure S5: **Verification and reproducibility (supplementary).** (A) MNIST verification: main seeds (42–46) yield $91.4 \pm 0.3\%$; held-out seeds (47–49) yield $89.8 \pm 0.9\%$, confirming generalization. (B) Figure-ground MNIST verification across the reported main and held-out settings. (C) HSIC weight ablation on figure-ground MNIST: moderate weights (0.01–0.1) perform best. The submitted revision controls evaluate HSIC weight 0 versus 0.01 on both seed sets.

The small numerical ε is used only for floating-point safety; analytically the leak conductance keeps the denominator positive. The total conductance and input resistance recorded for learning are

$$g_n^{\text{tot}} = 1 + U_n^E + U_n^I + G_n^{\text{den}}, \quad R_n^{\text{tot}} = (g_n^{\text{tot}})^{-1}.$$

In the additive control, inhibition is subtractive rather than shunting. With the same nonnegative conductance variables and nonnegative inputs, the implemented comparator is

$$V_n^{\text{add}} = U_n^E + D_n - U_n^I,$$

with optional additive-normalization controls reported separately in Fig. S9. Thus additive inhibition changes the signed voltage contribution and local additive eligibility, but it does not enter the denominator and does not gate conductance-stage path gain through R_n^{tot} .

Positive Parameterization and Raw-Parameter Gradients

Trainable synaptic and dendritic conductances are stored as unconstrained raw parameters θ and transformed by a positive map $g = \psi(\theta)$, with `softplus` used in the main reported runs:

$$\psi(\theta) = \log(1 + \exp \theta), \quad \psi'(\theta) = \sigma(\theta).$$

Equations in the main text give conductance-space gradients $\partial L / \partial g$. Before assigning gradients to the optimizer variables, the implementation applies the chain rule

$$\frac{\partial L}{\partial \theta} = \frac{\partial L}{\partial g} \psi'(\theta).$$

The exact-gradient reconstruction diagnostic includes this parameterization factor, as well as top- k activity masks and dendritic block-linear parameterization. This is why the reconstruction comparison is against raw autograd gradients rather than only against abstract conductance variables.

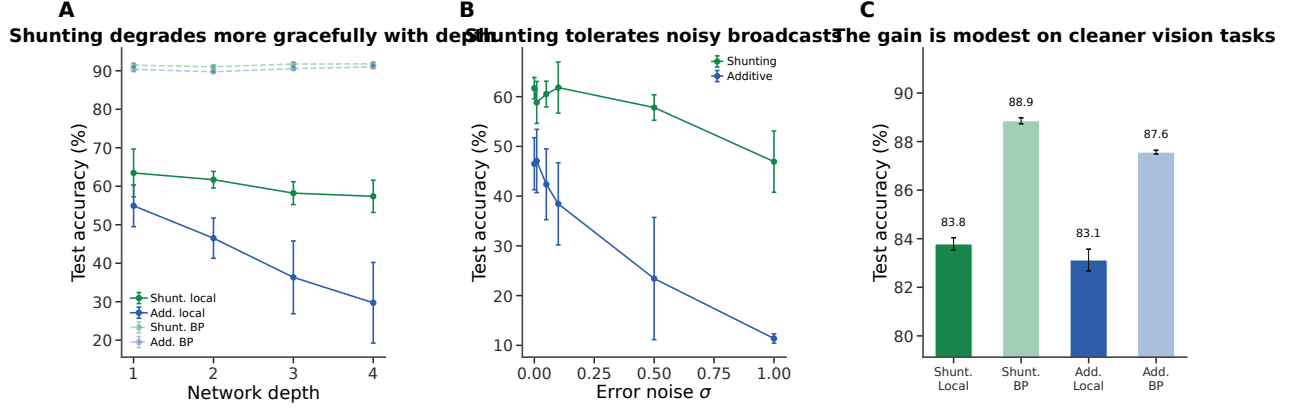


Figure S6: **Additional stress tests.** (A) Depth scaling: shunting local (green) degrades more gracefully from 63.5% to 57.4%; additive local (blue) drops from 54.9% to 29.7%. Backprop ceilings (dashed) remain stable. (B) Broadcast-noise robustness: shunting remains stable for $\sigma \leq 0.1$; additive degrades rapidly and reaches chance at $\sigma = 1$. (C) Fashion-MNIST: the shunting gain is modest on this cleaner benchmark (83.8% local vs. 88.9% backprop), consistent with the regime-dependent story in the main text.

Post-Voltage Reactivation

The conductance stage produces a pre-activation voltage V_n , after which each branch may transmit a monotone post-voltage activity $a_n = f_n(V_n)$. The main sweeps use the learnable bounded transform

$$f_n(V) = \frac{\tanh(m_n(V - b_n)) + 1}{2}, \quad m_n = \exp(\ell_{m,n}),$$

with derivative

$$f'_n(V) = \frac{m_n}{2} [1 - \tanh^2(m_n(V - b_n))].$$

When reactivation is enabled, exact path transport uses the effective gain $\tilde{\alpha}_n$: parent-to-child transport includes the parent activation derivative, and the local update converts activation-space error into pre-activation voltage error by multiplying by the child derivative. With identity transfer, this reduces to conductance-stage transport through α_n^{cond} .

Decoder and Error Coordinates

Let h denote the final soma/core activity passed to the decoder and let $\delta^y = \partial L / \partial \hat{y}$. For a linear decoder $\hat{y} = W_{\text{dec}} h$, the soma/core error is

$$\delta_0 = W_{\text{dec}}^\top \delta^y.$$

For a nonlinear decoder, used in the CIFAR-10 compact control ladder, the implementation uses the decoder-input Jacobian product

$$\delta_0 = J_{\text{dec}}(h)^\top \delta^y.$$

All LocalCA broadcast modes are defined in this soma/core error coordinate system. Decoder-update modes are: **backprop**, which uses autograd for decoder parameters; **local**, which is implemented for the cached linear readout and assigns

$$\hat{V}_{W_{\text{dec}}} L = \langle \delta^y h^\top \rangle_B, \quad \hat{V}_{b_{\text{dec}}} L = \langle \delta^y \rangle_B;$$

and **frozen**, which holds the decoder fixed. The local decoder gradient is output-space; δ_0 is the mapped soma/core error used by dendritic broadcasts.

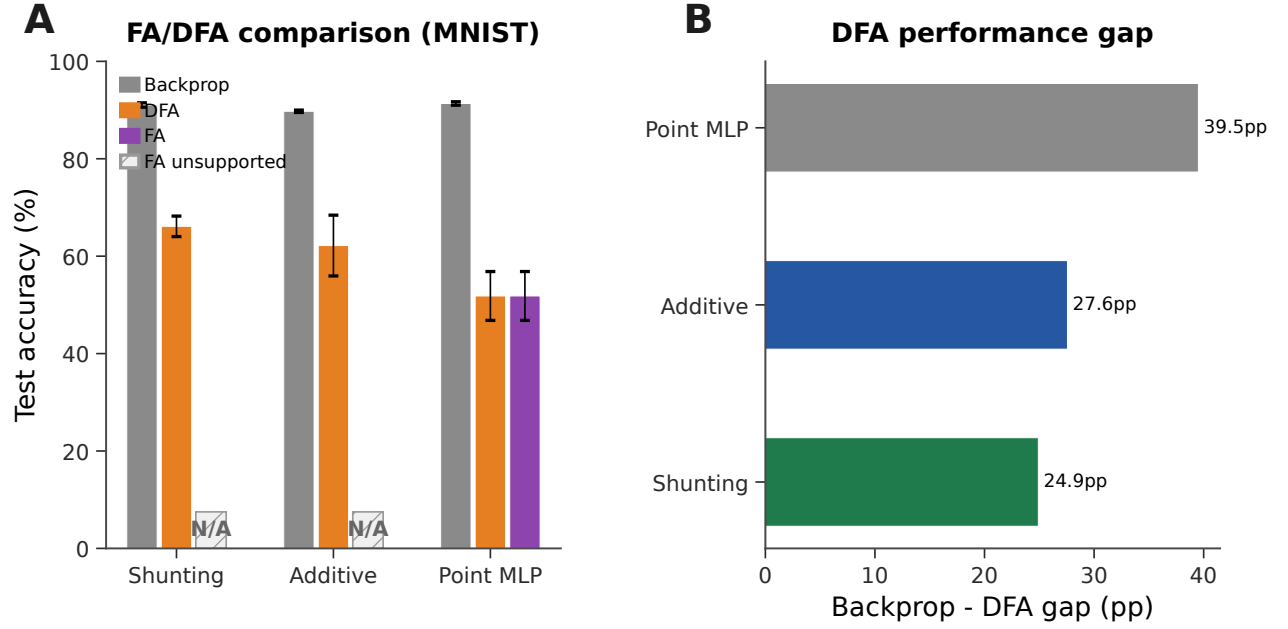


Figure S7: **Feedback alignment baselines (MNIST).** (A) Grouped comparison: standard backprop (neutral gray), DFA (orange), and FA (purple). Hatched N/A markers indicate that FA is not implemented for the block-structured dendritic architectures because the random feedback matrices are not dimensionally compatible with the tree-structured branch updates. DFA achieves 66.1% on shunting vs. 62.2% additive vs. 51.8% point MLP. (B) Backprop–DFA gap: dendritic architectures (24.9–27.6 pp) show smaller gaps than point MLPs (39.5 pp), suggesting conductance-based architecture is partially compatible with random feedback.

Biological Plausibility Assumptions

The model makes the following explicit assumptions. First, it uses the steady-state solution of the passive cable equation rather than temporal membrane dynamics; all path gains and gradient diagnostics refer to conductance-stage voltages at steady state. Second, each dendritic cell is a rooted tree, so each compartment has a unique path to the soma and the closed-form path gains α_n^{cond} and $\tilde{\alpha}_n$ are well-defined. Third, synaptic and dendritic conductance parameters are nonnegative by construction. Fourth, the main sweeps enforce nonnegative first-layer presynaptic drive through nonnegative inputs or a ReLU transfer stage. Fifth, post-voltage reactivation is handled through local $f'(V)$ factors along the path. Sixth, each dendritic synapse receives a restricted non-local feedback field–scalar, per-soma, low-rank, path-structured, or transported-oracle depending on the condition—while presynaptic activity, voltage, reversal potential, input resistance, covariance modulators, and confidence factors are compartment-local or slowly estimated local quantities.

Units and Parameterization

Table S12 summarizes the numerical conventions used throughout the implementation and diagnostics, including which quantities are constrained conductances and which are unconstrained readout parameters.

Representative Manuscript Architectures

Hyperparameters

Table S14 collects the core training hyperparameters for each main experiment. Learning rates are applied through parameter groups (top- k , dendritic block-linear, reactivation, decoder) with the values given in the representative config files; where a single LR is reported, every group uses that value.

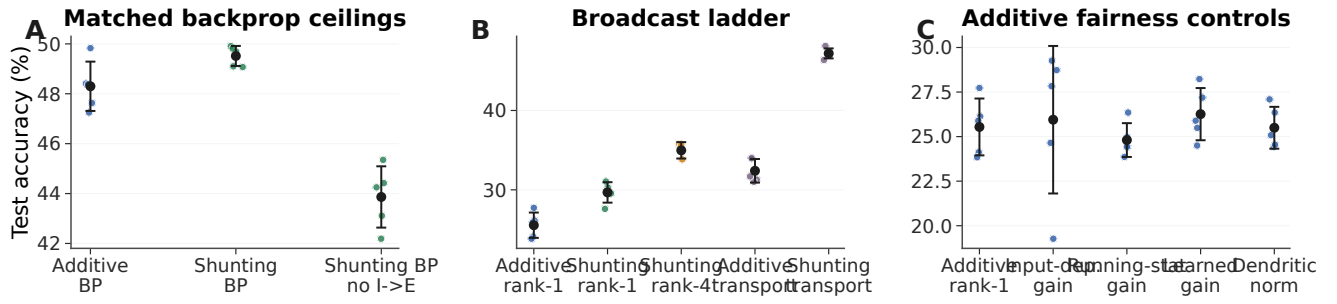


Figure S8: **CIFAR-10 control ladder in the compact direct-I-stream family.** (A) Matched backprop ceilings. With nonnegative inputs and a nonlinear decoder, shunting slightly exceeds additive under standard training, and removing learned I-to-E conductance lowers the shunting ceiling. (B) Broadcast ladder. Per-soma feedback remains weak on this harder dataset, random low-rank $K=4$ improves partially, and exact effective transported error nearly reaches the shunting backprop ceiling. (C) Additive fairness controls. Input-dependent gain, running-stat gain, learned gain, and dendritic normalization do not rescue additive per-soma feedback. This figure is a harder-data diagnostic, not a claim of competitive CIFAR-10 benchmarking.

Effective Broadcast Dimensionality in Main Architectures

Compute Resources

All experiments were run on an institutional GPU cluster. Reproduction runs use one NVIDIA A100 or H100-class GPU (40–80GB memory) per seed and do not require distributed training. Table S16 gives approximate single-GPU runtime ranges for the reported experiment families; unreported exploratory runs used additional cluster time and are not counted in the reproduction estimate.

Code, Data, and Asset Availability

MNIST, Fashion-MNIST, and CIFAR-10 are public datasets or public benchmark assets cited in the paper; we use them under their standard published access conditions and cite their original sources in the bibliography. To preserve double-blind review, the submission provides code through an anonymized supplementary ZIP rather than a public repository URL. The ZIP contains the core model and training code, selected diagnostic utilities, representative configuration files, tests, and reproduction instructions for the main experimental families; code and configurations will be de-anonymized after review. The paper does not release a new dataset or a stand-alone pretrained model asset. Dataset access pages are the original MNIST site / UCI entry (<https://archive.ics.uci.edu/dataset/683/mnist+database+of+handwritten+digits>), the Fashion-MNIST repository (<https://github.com/zalandoresearch/fashion-mnist>), and the CIFAR page (<https://www.cs.toronto.edu/~kriz/cifar.html>).

LocalCA Broadcast and Gradient Assignment

LocalCA first maps the output loss derivative to the soma/core error δ_0 and then constructs a branch error field. In the main reported `per_soma` mode, if the layer width matches the soma/core error dimension, each soma-specific error component is broadcast to the corresponding dendritic compartments:

$$e_{n,b}^a = \delta_{0,b} \in \mathbb{R}^{d_0}.$$

The stricter scalar control compresses the soma/core error to one value per example,

$$\bar{\delta}_b = \frac{1}{d_0} \sum_{c=1}^{d_0} \delta_{0,bc}, \quad e_{n,b}^a = \bar{\delta}_b \mathbf{1}_n,$$

Condition	Core	Training / broadcast	I-to-E	Test acc.
standard	additive	backprop	yes	48.3 ± 1.0
standard	shunting	backprop	yes	49.5 ± 0.4
standard, no learned inhibition	shunting	backprop	no	43.9 ± 1.2
per-soma	additive	5F per-soma	yes	25.5 ± 1.6
per-soma, input-dependent gain	additive	5F per-soma	yes	25.9 ± 4.1
per-soma, running-stat gain	additive	5F per-soma	yes	24.8 ± 0.9
per-soma, learned gain	additive	5F per-soma	yes	26.3 ± 1.5
per-soma, dendritic normalization	additive	5F per-soma	yes	25.5 ± 1.2
per-soma	shunting	5F per-soma	yes	29.7 ± 1.3
per-soma, no learned inhibition	shunting	5F per-soma	no	22.3 ± 0.7
transported error	additive	5F path transport	yes	32.4 ± 1.5
rank-4	shunting	5F low-rank	yes	35.0 ± 1.0
transported error	shunting	5F path transport	yes	47.2 ± 0.6
transported error, no learned inhibition	shunting	5F path transport	no	38.6 ± 1.0

Table S6: **Exact CIFAR-10 values for the 70-run control ladder (5 seeds per condition, validation-best epoch).**

All rows use flattened CIFAR-10 in $[0, 1]$, nonnegative transfer outputs, positive conductance transforms, a compact depth-4 dendritic core, and a nonlinear decoder. LocalCA maps output error back to decoder-input/soma coordinates through the decoder Jacobian. The useful conclusions are narrow but important: learned shunting slightly improves the matched backprop ceiling, learned I-to-E conductance is necessary for the strongest shunting results, transported error nearly closes the shunting LocalCA gap, and additive per-soma fairness controls do not explain the transported-shunting result.

where d_0 is the dimension of the soma/core error. If dimensions do not match, the implementation falls back to scalar-style expansion rather than silently inventing an unmatched per-soma map. The low-rank controls use fixed random projection and mixing matrices P_K and Q_n ,

$$c_b = P_K \delta_{0,b} \in \mathbb{R}^K, \quad e_{n,b}^a = Q_n c_b.$$

The pathway-vector controls use router-inferred pathway roles to gate and transport a low-rank role vector. The transported oracle uses the effective tree recursion. If p is the parent of child c , activation-space error is propagated by

$$e_c^a = e_p^a f'_p(V_p) R_p^{\text{tot}} g_{c \rightarrow p}^{\text{den}},$$

and the local pre-reactivation voltage error used in conductance eligibility is

$$e_n^V = \begin{cases} e_n^a f'_n(V_n), & \text{post-voltage reactivation enabled,} \\ e_n^a, & \text{identity transfer.} \end{cases}$$

This operational definition is the implementation counterpart of the effective path gain $\tilde{\alpha}_n$ in Theorem 1.

For shunting branches, the conductance-space LocalCA gradients assigned before raw-parameter transformation are

$$\hat{\nabla}_{g_{nj}^E} L = \langle e_n^V x_{nj}^E R_n^{\text{tot}} (1 - V_n) \rangle_B, \quad \hat{\nabla}_{g_{nj}^I} L = \langle e_n^V x_{nj}^I R_n^{\text{tot}} (0 - V_n) \rangle_B,$$

and for a dendritic coupling from child c to parent n ,

$$\hat{\nabla}_{g_{c \rightarrow n}^{\text{den}}} L = \langle e_n^V R_n^{\text{tot}} (a_c - V_n) \rangle_B.$$

For additive controls, the matching local derivatives are

$$\hat{\nabla}_{g_{nj}^E} L = \langle e_n^V x_{nj}^E \rangle_B, \quad \hat{\nabla}_{g_{nj}^I} L = -\langle e_n^V x_{nj}^I \rangle_B, \quad \hat{\nabla}_{g_{c \rightarrow n}^{\text{den}}} L = \langle e_n^V a_c \rangle_B.$$

The optimizer applies the usual descent step; equivalently, the sign can be absorbed into the definition of the broadcast error. After these conductance-space gradients are formed, they are multiplied by the positive-transform derivative $\psi'(\theta)$ and by any active synapse masks before assignment to raw parameters.

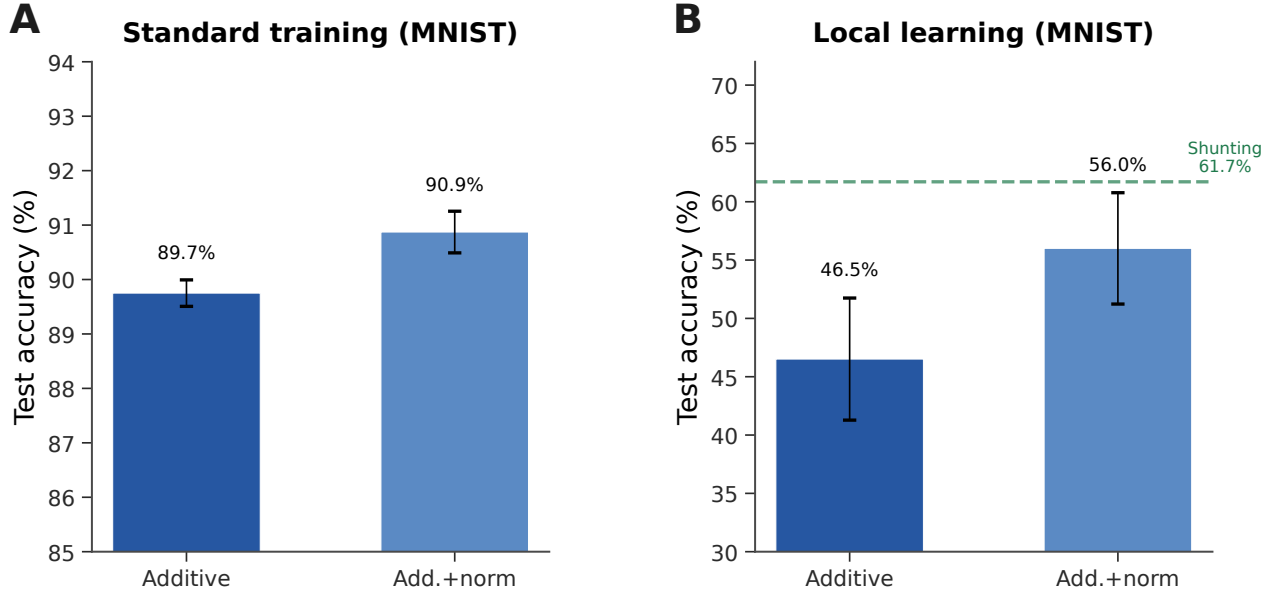


Figure S9: **Additive + normalization control (MNIST stress-regime diagnostic)**. **(A)** Standard backprop: normalization provides a small boost (89.7% $\rightarrow$ 90.9%). **(B)** Local learning: normalization improves additive from 46.5% to 56.0% (+9.5 pp), partially closing the gap to shunting (61.7%, dashed green). These lower local accuracies come from a separate stress-regime diagnostic, not from the capacity-calibrated competence setting in Table S5.

4F and 5F Local Modulators

The 3F rule uses only the local eligibility and broadcast error above. The 4F and 5F variants multiply the conductance-space gradient by local reliability factors before raw-parameter transformation. The 4F factor is an online branch-soma covariance proxy,

$$r_n^{4F} = \frac{\text{Cov}(\bar{V}_n, \bar{V}_0)}{\sqrt{\text{Var}(\bar{V}_n)\text{Var}(\bar{V}_0)} + \epsilon}.$$

In the implementation this proxy is clamped to the positive stability range $[0.1, 2.0]$ before it multiplies the local gradient; sign-opposed covariance therefore reduces the multiplier rather than reversing update direction. The 5F factor adds a confidence term based on how much branch voltage variance remains predictable from parent or soma-level activity:

$$\phi_n = \text{clip}_{[0.25, 4.0]} \left(\frac{\text{Var}(V_n)}{\sigma_{\text{res},n}^2 + \epsilon} \right),$$

where $\sigma_{\text{res},n}^2$ is estimated online by the configured residual-variance estimator. In the main 5F runs, the multiplier is $r_n^{4F} \phi_n$. The 5F sensitivity diagnostic in Fig. S2 varies the clamp and EMA rate to verify that the reported MNIST performance is not a single brittle clamp setting.

Algorithm

Algorithm 1 gives the update order for the main 5F per-soma LocalCA condition and shows where scalar, higher-bandwidth, and transported-error controls enter.

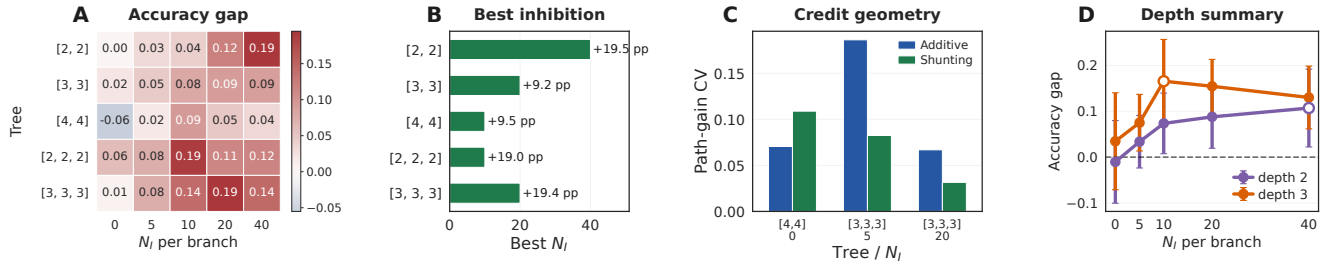


Figure S10: **Morphology shifts the useful inhibitory regime.** (A) Shunting-minus-additive LocalCA accuracy gap across dendritic morphologies and inhibitory synapse counts. (B) Best inhibitory count for each morphology. (C) Representative path-gain diagnostic for selected morphology cells, used to connect the performance map to credit geometry. (D) Average shunting advantage by depth, with markers showing the peak inhibitory count for each depth.

Algorithm 1 Main 5F Per-Soma LocalCA Update

- 1: **Input:** Dendritic model, batch (x, y) , optimizer
 - 2: Forward pass; loss L , output error δ^y
 - 3: Somatic/core error $\delta_0 = \text{decoder-input Jacobian product } J_{\text{dec}}(h_{\text{core}})^\top \delta^y$ (or $W_{\text{dec}}^\top \delta^y$ for a linear decoder)
 - 4: **for** each layer n (reverse) **do**
 - 5: Per-soma feedback $e_n = \delta_0$ when dimensions match; scalar/low-rank/path controls replace this line
 - 6: Update EMA estimates for r_n^{AF} and ϕ_n from batch voltages
 - 7: Convert to pre-activation voltage error e_n^V using the local activation derivative when needed
 - 8: $\widehat{V}_{g_j^{\text{syn}}} L \leftarrow r_n^{\text{AF}} \phi_n \langle x_j R_n^{\text{tot}} (E_j - V_n) e_n^V \rangle_B$
 - 9: $\widehat{V}_{g_{c \rightarrow n}^{\text{den}}} L \leftarrow r_n^{\text{AF}} \phi_n \langle R_n^{\text{tot}} (a_c - V_n) e_n^V \rangle_B$
 - 10: Map conductance gradients to raw parameters by multiplying by $\partial g / \partial \theta$
 - 11: **end for**
 - 12: Optional controls replace line 5 with scalar, rank- K , path-structured, or transported-oracle feedback.
 - 13: Clip gradients; optimizer step
-

C Theoretical Details

This section collects theory-adjacent material that supports implementation and interpretation but is not part of the main proof. The central derivation remains Theorem 1, Corollary 1, and Prop. 2.

Random-Broadcast Alignment Intuition

Feedback-alignment arguments suggest that random or low-rank feedback can be useful when the induced local update remains positively correlated with the exact gradient [13]. In this model, that correlation depends on how local eligibility factors co-vary with the conductance-stage path gain in Eq. (4). We treat this as intuition only. The central theoretical object in the paper is still the exact pre-activation compartment error $\partial L / \partial V_n$; under identity upward transfer this reduces to $\alpha_n^{\text{cond}} \delta_0$, while post-voltage activation uses $\tilde{\alpha}_n \delta_0$. The empirical question is how faithfully different broadcast fields approximate that quantity.

Morphology-Aware Extensions

The following variants are implemented controls or architectural extensions used to test whether coarse morphology-aware feedback can approximate exact path transport. They are not required for the main 5F per-soma results, but they define the path-propagation, depth, normalization, and pathway-vector controls reported in the appendix figures and tables.

Inhibitory input mode	Training / broadcast	Test accuracy
Direct inhibitory stream	LocalCA, per-soma feedback	0.852 ± 0.007
Direct inhibitory stream	LocalCA, exact effective path transport	0.952 ± 0.000
Explicit I cells	LocalCA, path transport, train I-cell dendrites	0.942 ± 0.002
Explicit I cells	LocalCA, path transport, freeze I-cell dendrites	0.938 ± 0.002
Explicit I cells	standard backprop	0.951 ± 0.003

Table S7: **One-feedforward-layer input-mode path-gain probe on noise resilience (3 seeds).** All rows use one excitatory dendritic population with $[3, 3]$ morphology and nonnegative inputs. The direct-stream rows have no explicit inhibitory neurons; the input stream drives learned branch-level inhibitory conductance banks directly. The explicit-I rows use a matched inhibitory population with $[3, 3]$ morphology. Exact effective path transport closes the direct-stream per-soma gap, and explicit inhibitory cells nearly match both direct-stream path transport and the explicit-I backprop ceiling.

Path-integrated propagation. Modulate broadcast error by $\pi_n = \pi_{n-1} \cdot R_{n-1}^{\text{tot}} \cdot \bar{g}_{n-1}^{\text{den}}$, approximating depth attenuation from Eq. (4) without computing the exact sample-specific path gain.

Depth modulation. Per-branch scaling $\kappa_j = \kappa_{\text{base}} / (d_j + d_0)$, mirroring cable attenuation and testing whether a simple depth prior can stabilize distal updates.

Dendritic normalization. $\Delta g_j^{\text{den}} \leftarrow \Delta g_j^{\text{den}} / (\sum_k g_k^{\text{den}} + \epsilon)$, analogous to homeostatic scaling [25] and used as an additive-control normalization comparison.

Pathway-vector feedback. For tasks with latent pathway structure, the broadcast becomes a role vector inferred from the router. Local pathway activity gates this vector. Upward block transport then passes it to earlier branches, aligning their feedback with downstream descendants.

Router-derived branch roles. Each branch receives a role profile inferred from router assignments and incoming block weights rather than from a hand-coded apical/basal label. Plasticity is then modulated by branch selectivity and a synapse-specific role-alignment factor, emphasizing pathway-consistent updates without imposing a heuristic branch taxonomy.

HSIC Auxiliary Objectives

Following [23], we use kernel-based HSIC objectives:

$$\mathcal{L}^{\text{self}} = B^{-2} \text{tr}(\mathbf{K}_Z \mathbf{H} \mathbf{K}_Z \mathbf{H}), \quad \mathcal{L}^{\text{target}} = -B^{-2} \text{tr}(\mathbf{K}_Z \mathbf{H} \mathbf{K}_Y \mathbf{H}).$$

Moderate weights (0.01–0.1) help on figure-ground MNIST; negligible on MNIST. Online statistics (r_n^{4F} , ϕ_n) use Welford’s algorithm [24].

D Task Construction and Robustness Protocols

This section specifies how the benchmark and synthetic tasks are constructed, including which tasks preserve the nonnegative-input regime assumed by the conductance model. It also records the stress-test protocols used for depth and broadcast-noise robustness.

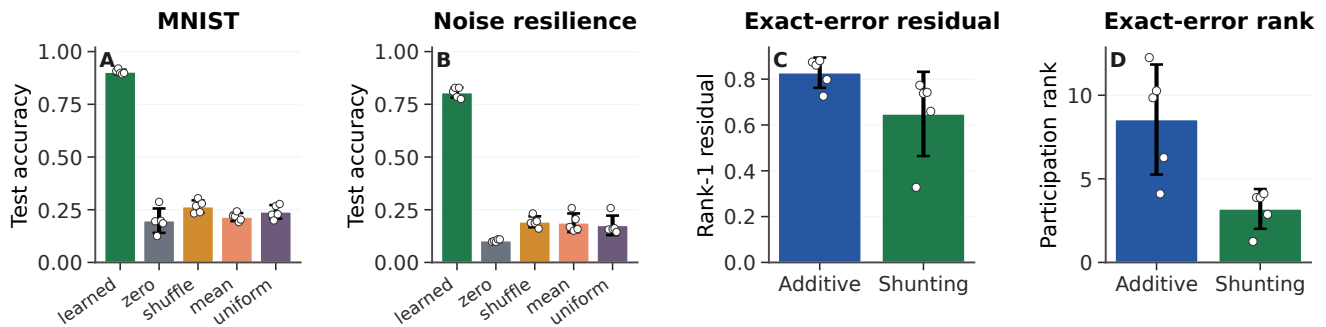


Figure S11: **Post-training inhibition intervention and exact-error compressibility.** (A,B) Post-training inhibitory-conductance interventions on trained shunting LocalCA checkpoints. Accuracy is measured on a 512-example held-out subset; dots show seeds. Removing, shuffling, or clamping learned inhibition collapses performance, showing that the learned inhibitory field is sample-dependent and functionally important. (C,D) SVD diagnostics of the exact compartment-error matrix on matched MNIST $N_I=5$ runs. Shunting has a smaller unrestricted rank-1 residual and lower participation rank than additive, so the exact error field is more intrinsically compressible. Error bars are s.d. across 5 seeds.

General Data Handling

MNIST, Fashion-MNIST, and CIFAR-10 are used through standard public dataset loaders. Images are flattened before entering the dendritic core, scaled to $[0, 1]$, and passed through the configured nonnegative transfer stage in the main experiments. MNIST-derived synthetic tasks inherit the flattened 28×28 input geometry. Random task components, such as corruption projections and context masks, are generated from fixed seeds and reused across the corresponding train, validation, and test splits. Reported validation-best results use validation accuracy for checkpoint selection and test accuracy for final reporting.

The paper does not use data augmentation as a performance device. CIFAR-10 is intentionally a flattened harder-data diagnostic under the same conductance constraints rather than a competitive vision benchmark.

Supervised Benchmark Tasks

The supervised tasks below provide the capacity-calibrated classification setting, the harder-data diagnostic, and the structured nonnegative synthetic tasks used to probe broadcast limitations.

MNIST and Fashion-MNIST. The standard classification tasks use ten output classes and nonnegative pixel inputs. They serve as capacity-calibrated tests of whether per-soma 5F LocalCA can approach a matched backpropagation reference on clean supervised data. Fashion-MNIST is included as a modest distribution shift with the same input and output dimensionality as MNIST.

CIFAR-10 harder-data diagnostic. CIFAR-10 inputs are flattened RGB images in $[0, 1]$. The compact CIFAR family uses a smaller dendritic core with a nonlinear decoder and decoder-aware LocalCA error mapping. We use this setting to test whether the broadcast-fidelity ladder persists on a harder data distribution: per-soma feedback is weak, higher-rank feedback helps, and exact effective path transport approaches the matched shunting backprop ceiling.

Figure-ground MNIST. Inputs are figure-ground MNIST examples used for 10-way digit classification. The right half of the image carries the original MNIST digit, while the left half is replaced by low-amplitude uniform noise; all pixels therefore remain nonnegative. The repository configuration for this task is historically named `context_gating`, but the reported task does not require a context-dependent label rule. It tests whether credit assignment can exploit a simple spatial signal/distractor structure without leaving the nonnegative-drive regime of the

Model implication	Biological prediction
Path-local inhibitory conductance changes α_n^{cond} only for descendants whose credit path crosses the inhibited compartment.	Branch-local inhibitory perturbation should distort plasticity or teaching-signal efficacy for synapses below the perturbed compartment more than for sister branches.
The log derivative $\partial \log \alpha_n^{\text{cond}} / \partial G_k^I = -R_k^{\text{tot}}$ predicts stronger credit gating in high-resistance compartments.	The same inhibitory conductance change should have larger effects on low-conductance/high-resistance branches than on already high-conductance branches.
Rank-1 broadcast succeeds when the exact compartment-error matrix is low-rank and fails when branch identity matters.	Tasks requiring route- or context-specific branch credit should need richer dendritic teaching signals than a single scalar modulatory broadcast.

Table S8: **Experimental predictions from the path-gain mechanism.** These are qualitative biological predictions, not claims tested directly in this paper.

conductance model. The HSIC auxiliary objective (weight 0.01) is used in the main LocalCA competence runs to decorrelate layer representations, adding roughly 3 pp under local learning in the ablation.

Noise resilience. Inputs are flattened MNIST images corrupted by structured channel noise: a fixed 50-dimensional latent Gaussian perturbation is projected into input space and added to each example, with $\sigma_{\text{task}} = 1.5$, then clipped to $[0, 1]$. The projection is fixed across training, so the network must filter correlated interference to recover the underlying class signal. Credit assignment must remain coherent under the corrupted signal, making this task particularly sensitive to path-gain dispersion.

Cue integration. Two noisy cue streams (A and B) are presented simultaneously, each carrying partial information about one of 2 classes. A binary one-hot context signal indicates which cue is reliable on each trial (cue A or cue B), and each cue vector is clipped to $[0, 1]$ after noise is added. The fixed-pathway variant duplicates the context into both cue branches; the learned-routing variant requires the router to discover the cue separation from data. This task is used in the cue-routing feedback-rank diagnostic to test structured pathway-vector broadcast while staying inside the nonnegative-input regime.

Depth Scaling and Noise Robustness

These robustness protocols are used only as stress diagnostics. They keep the same broad model family while changing dendritic depth or corrupting the broadcast signal.

Depth scaling. Varying dendritic depth from 1–4 layers (branch factors $[9]$ to $[3, 3, 3, 3]$) tests whether local credit deteriorates with longer paths. Shunting local degrades from 63.5% to 57.4%; additive local degrades more steeply from 54.9% to 29.7%. The shunting advantage grows with depth ($+8.5 \rightarrow +27.7$ pp). Backprop ceilings remain stable at about 90–92%. See Fig. S6A.

Noise robustness. Gaussian noise $\mathcal{N}(0, \sigma^2)$ is added to the broadcast error to test whether learning depends on a finely tuned scalar magnitude. Shunting is robust to $\sigma \leq 0.1$ (about 62%) while additive drops from 46.5% to chance at $\sigma=1.0$, confirming that shunting credit signals carry useful learning information beyond the broadcast magnitude. See Fig. S6B.

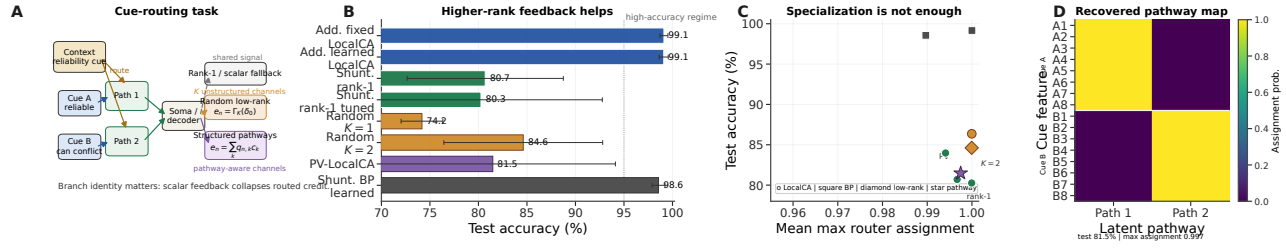


Figure S12: **Structured feedback when restricted feedback fails.** (A) Cue-routing task schematic. Context determines which noisy cue is reliable on each trial, so different dendritic pathways should receive different teaching signals. (B) Five-seed comparison of per-soma, random low-rank, and pathway-structured feedback. Moving beyond the per-soma / single-channel baselines helps, while the best structured repair remains open. (C) Test accuracy vs. router specialization across learned-router conditions. High specialization alone does not guarantee success. (D) Recovered pathway assignments for a representative learned-router run. Cue-A and cue-B features segregate into separate latent pathways even when feedback quality, rather than router discreteness, is the limiting factor.

Setting	Broadcast	Test accuracy
<i>Noise resilience, shunting, rank bridge</i>		
per-soma	per-soma feedback	0.662 ± 0.016
path propagation	recursive attenuation proxy	0.762 ± 0.047
low-rank	$K=1$	0.453 ± 0.126
low-rank	$K=2$	0.630 ± 0.042
low-rank	$K=4$	0.764 ± 0.025
low-rank	$K=8$	0.795 ± 0.019
path transport	exact effective path transport	0.847 ± 0.018
<i>Cue routing, learned shunting router, rank/structure sweep</i>		
per-soma	tuned per-soma feedback	0.803 ± 0.125
low-rank	$K=1$	0.742 ± 0.022
low-rank	$K=2$	0.846 ± 0.082
low-rank	$K=4$	0.748 ± 0.085
pathway-vector	tuned structured rank-2	0.815 ± 0.126

Table S9: **Rank bridge and rank-structure comparison.** On shunting noise resilience, both recursive path propagation and higher-rank random low-rank broadcast close a substantial fraction of the gap between per-soma feedback and exact effective path transport, with low-rank continuing to improve up to $K=8$. On cue routing, the learned-router sweep shows that moving beyond the per-soma baseline helps, but the best unstructured low-rank setting ($K=2$) is still slightly better than the tuned structured pathway-vector variant. We therefore interpret routed credit assignment as an open higher-rank problem rather than as a solved pathway-vector rescue.

Paper object	Code anchor	Role
Shunting branch voltage	<code>DendriticBranchLayer.forward</code> with <code>use_shunting=True</code>	Implements numerator/denominator conductance integration and stores total conductance for diagnostics.
Additive control voltage	<code>DendriticBranchLayer.forward</code> with <code>use_shunting=False</code> ; <code>normalize_additive_voltage</code> for the normalization control	Uses the same branch tree and synapse banks, but inhibitory input is subtractive rather than a denominator load.
Broadcast-mode switch	<code>LocalCreditAssignment._apply_local_rule_gradients</code> ; <code>LocalRuleConfig.error_broadcast_mode</code>	Chooses scalar/per-soma, local-mismatch, low-rank, path-transport, and pathway-vector feedback fields.
Low-rank feedback	<code>LocalCreditAssignment._compute_low_rank_broadcast</code> ; <code>broadcast_rank</code>	Fixed random low-rank broadcast controls used in the bandwidth ladder.
Path-transport oracle	<code>LocalCreditAssignment._precompute_path_transport_errors</code>	[C] transported-error upper bound using the effective tree recursion.
Path-propagation proxy	<code>LocalCreditAssignment._precompute_path_propagation_factors</code>	Approximate morphology-aware attenuation control, distinct from exact path transport.
Shunting eligibility	<code>LocalCreditAssignment._apply_local_rule_gradients</code> conductance branch	Assigns $xR^{\text{tot}}(E - V)e$ gradients, then multiplies by the positive-transform derivative.
Additive eligibility	<code>LocalCreditAssignment._apply_local_rule_gradients</code> additive branch	Assigns the matched signed additive derivative, with optional additive-gain controls.
4F / 5F reliability gates	<code>_compute_layer_rho</code> , <code>_compute_layer_phi*</code> , <code>clamp_phi</code> , <code>FiveFactorConfig</code>	Slow reliability preconditioners used by 4F/5F, not additional fast task-error channels.
Experiment knobs	<code>LocalRuleConfig</code> ; YAML fields <code>rule_variant</code> , <code>error_broadcast_mode</code> , <code>broadcast_rank</code> , <code>five_factor</code>	Reproduces the rule family, feedback bandwidth, and 5F stabilizer settings.

Table S10: **Equation-to-code map.** Code anchors are symbolic names rather than line numbers so that minor refactors do not invalidate the reproducibility map.

Parameter family	Forward role	Update source	Feedback / auxiliary signal
Excitatory and inhibitory synaptic conductances	Branch E/I drive	LocalCA conductance eligibility	Scalar, per-soma, low-rank, path-structured, or transported feedback plus local presynaptic drive and voltage
Dendritic coupling conductances	Child-to-parent tree transfer	LocalCA conductance eligibility	Same branch feedback field, child activity, parent voltage, and input resistance
Reactivation slope/bias	Post-voltage transfer $f(V)$	Configured local/autograd schedule	Local activation derivative and available batch objective gradients; not part of the conductance-factorization theorem
Top- k synapse banks	Sparse input selection and selected weights	Masked LocalCA on selected banks; inactive updates only when enabled	Same branch feedback field, multiplied by top- k masks and local modulators
Linear decoder	Readout from final core activity	Local output-space decoder gradient or autograd	δ^y for local decoder mode; autograd for backprop mode
Nonlinear decoder	Compact CIFAR readout	Autograd for decoder parameters; Jacobian product for dendritic error mapping	$J_{\text{dec}}(h)^\top \delta^y$ maps output error into soma/core coordinates
HSIC auxiliary	Batch dependence regularizer	Auxiliary gradient on selected local representations when configured	Batch-global kernel gradient; reported as an auxiliary objective, not a local theorem term
Router / pathway roles	Structured cue-routing pathway selection	Configuration-dependent router/pathway updates	Pathway-vector controls use router-derived role signals; this remains an exploratory higher-rank mechanism

Table S11: **Update scope by parameter family.** The exact factorization covers conductance-stage branch and dendritic-coupling parameters. Decoder, reactivation, HSIC, and router mechanisms are implementation-level training components and are therefore documented separately from the theorem.

Quantity	Symbol	Convention
Voltage	V	Conductance-stage shunting voltage in $[0, 1]$ under nonnegative drive; additive controls may be signed
Conductances	$g^{\text{syn}}, g^{\text{den}}$	Nonnegative via $g = \text{softplus}(\theta)$; local gradients are mapped to raw θ by the transform derivative
Leak conductance	g^{leak}	Set to 1
Input resistance	R^{tot}	≤ 1

Table S12: Units and normalization.

Setting	Encoder	E layers	Tree	Synapses / branch	Seeds	Notes
Gradient fidelity / path-gain / noise-resilience mechanism	id.+ReLU	[128, 128]	[3, 3]	$N_E=40$, $N_I \in \{0, 5, 10, 20, 40\}$	5 fidelity, 5 oracle	Main mechanism sweeps in Figs. 2 and 3; reactivation <code>param_tanh</code> .
MNIST / figure-ground local competence	id.+ReLU	[128]	[3, 3]	$N_E=40$, $N_I=20$	5	Best local 5F per-soma runs in Table S5; reactivation <code>param_tanh</code> .
Fashion-MNIST competence	id.+ReLU	[128]	[3, 3]	$N_E=40$, $N_I=20$	5	Dedicated five-seed standard-vs-local competence sweep; reactivation <code>param_tanh</code> .
Cue-routing PV-LocalCA	router	[64]	[2]	$N_E=10$, $N_I=4$	5	Learned router with two pathway groups and rank-2 structured feedback; reactivation <code>param_tanh</code> .
CIFAR-10 compact direct-I-stream control ladder	id.+ReLU	[20] E, no I cells	[3, 3, 3, 3]	$N_E=25$, $N_I=25$ direct I-to-E, with matched no-I controls	5 per condition	Harder-data extension with input-driven inhibitory conductance, decoder [32, 16], decoder-aware soma mapping, and additive fairness controls.
Morphology $\times$ inhibition appendix map	id.+ReLU	[128, 128]	[2, 2], [3, 3], [4, 4], [2, 2, 2], [3, 3, 3]	$N_E=40$, $N_I \in \{0, 5, 10, 20, 40\}$	3	Supportive 3-seed sweep; reactivation <code>param_tanh</code> .

Table S13: **Representative architectures used in the manuscript.** All dendritic rows use explicit branch-level excitatory and inhibitory synapse banks without direct somatic synapses, nonnegative conductances, and a nonnegative first-layer drive enforced by an explicit ReLU transfer stage in the main runs. The tree column gives dendritic morphology per excitatory neuron, and the synapse counts report excitatory and inhibitory synapses per branch.

Setting	Opt.	LR	Batch	Epochs	Weight decay	Notes
MNIST / Fashion-MNIST / figure-ground MNIST (5F per-soma LocalCA)	Adam	10^{-3} / $5 \cdot 10^{-4}$ (block / react.)	256	100	0	fixed-epoch, no early stopping; <code>param_tanh</code> reactivation
MNIST / Fashion-MNIST / figure-ground MNIST (BP ceiling)	Adam	10^{-3} / $5 \cdot 10^{-4}$	256	100	0	matched to LocalCA setup
Gradient fidelity / path-gain / noise resilience	Adam	10^{-3}	256	50	0	5 seeds, hooks enabled for diagnostic capture
Morphology $\times$ inhibition regime map (supportive)	Adam	10^{-3}	256	50	0	3 seeds
Cue routing	Adam	10^{-3}	256	90	0	early stopping with patience 30
CIFAR-10 compact direct-I-stream depth-4 LocalCA	Adam	10^{-3} / 10^{-4} (block / react.)	256	400	0 / 0.01	grad-clip 5.0, early stop patience 50, nonlinear decoder
CIFAR-10 compact direct-I-stream depth-4 BP	Adam	10^{-3}	256	200	0.01	early stop patience 40, matched ceiling

Table S14: **Training hyperparameters by experiment.** All runs use Adam with default $\beta_1=0.9$, $\beta_2=0.999$, $\epsilon=10^{-8}$. Reactivation and block-linear parameter groups use a slower LR to keep conductance-level dynamics stable. Representative configs are provided in the anonymized supplementary ZIP.

Setting	Soma/core dim.	Output classes	Main feedback field
Gradient fidelity / path-gain / noise resilience	128, 128	10	Per-soma soma/core error at matched-width dendritic layers
MNIST / Fashion-MNIST competence	128	10	Per-soma soma/core error
Figure-ground MNIST competence	128	10	Per-soma soma/core error
Cue routing	64	2	Tuned per-soma baseline; higher-rank tests use low-rank or pathway-vector feedback
Compact CIFAR-10 harder-data extension	20	10	Vector-valued per-soma after decoder-aware mapping; higher-rank tests use explicit low-rank or path transport

Table S15: **Effective dimensionality of the main feedback in the reported architectures.** Output loss gradients are first mapped into decoder-input/soma coordinates. The main competence and mechanism runs use the implemented per_soma mode when the soma/core dimension matches layer width; scalar, low-rank, pathway-vector, and transported-error conditions are labeled separately as controls or oracles.

Experiment family	Runs	Time / run	Approx. GPU-h
Competence and verification	~25	5–20 min	4–9
Gradient, path-gain, inhibition, and oracle diagnostics	~80	10–30 min	15–40
Morphology, stress, rule, and feedback controls	~120	5–35 min	20–70
CIFAR-10 control ladder	70	0.8–2.5 h	60–175
Figure generation and CPU-side summaries	–	< 5 h total	< 5

Table S16: **Approximate compute for reported results.** Ranges reflect single-seed wall-clock variation across A100/H100-class GPUs, data-loading overhead, and early stopping.

Asset	Use in this paper	Access / license or terms note
MNIST	Digit classification and derived nonnegative synthetic tasks	Public benchmark from the original MNIST site / UCI entry; UCI lists DOI 10.24432/C53K8Q and asks users to follow the original acknowledgement policy; no raw-data redistribution.
Fashion-MNIST	Apparel classification stress test	Public Zalando Research benchmark; repository is MIT licensed; no raw-data redistribution beyond standard dataset loaders.
CIFAR-10	Flattened harder-data diagnostic	Public CIFAR dataset site asks users to cite Krizhevsky’s technical report; no raw-data redistribution.
Synthetic tasks	Figure-ground MNIST, noise resilience, cue integration	Generated procedurally from public benchmarks or random seeds described in Appendix D; no new third-party asset is introduced.

Table S17: **Existing assets used in the manuscript.** We credit the original sources in the bibliography, use public benchmark assets under their published access conditions, and do not redistribute third-party raw data in the anonymous submission.